Reconstruction Buys Variance, Not Independence: A Tokenizer Keeps the Microstructure That Duplicates OHLCV and Drops the Signed Channels

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Abstract

Two-stage models for financial time series learn a tokenizer under a reconstruction objective and then fit an autoregressive model to the resulting discrete codes. This design carries an unstated assumption: that a representation good enough to reconstruct from is good enough to forecast from. We show that it fails in a specific, predictable direction. Reconstruction objectives allocate code capacity by variance and covariance and never by downstream value, so wherever the budget is tight and the competing dimensions are correlated enough to amortize one another, a low-variance channel that is weakly covariant with price is excluded deterministically rather than by chance. That is the statistical profile of trade-flow microstructure against price.

We establish the mechanism on an entropy-calibrated fixture, where a planted signal of known information content is lost to tokenization while the same quantity planted directly in token space is recovered by the identical backbone — localizing the loss to the tokenizer–backbone interface, not to model capacity. Re-specifying that interface so the fine subtoken encodes the current bar alone, and adding a microstructure class weight, restores per-bar legibility on the fixture at unchanged bits per token. We then convert that property into a pre-registered hard stop on real data.

On real market data the gate fired, and it named the channels. 97.3% of the shortfall falls on trade-flow imbalance and signed count imbalance — the two *signed* channels — while two of the four magnitude channels clear the gate outright and the other two miss it by 0.0025 and 0.0038; under a shuffled-microstructure placebo the signed channels collapse toward their base rates while the magnitude channels hold, and reading the whole 512-bar window rather than one bar's token recovers nothing. **The tokenizer preserves the microstructure that duplicates what price already carries and loses the microstructure that is independent of it — which is the part a microstructure model would want.** That split was not anticipated, and it is the paper's result.

The firing was pre-registered as the mechanism's own modal prediction, so the five-cell trading ablation it gates was never run and this paper reports no economic outcome. The finding is bounded by one budget point, one objective and one architecture: it establishes that this tokenizer does not carry the signed channels, not that none can.

1 Introduction

A tokenizer is a lossy compressor, and something has to decide what it discards. In the two-stage design now common for time series — a tokenizer trained to reconstruct, then an autoregressive model fitted to the resulting discrete codes — that decision is made by a reconstruction objective, and the forecaster that consumes the codes has no vote in it. The arrangement is inherited from image and audio tokenization, where it is sound: there, reconstruction *is* the task, so an objective that preserves what reconstructs well preserves what matters. Carried into forecasting, the same objective is being asked a question it was never given: not how much of the input survives, but *which parts*.

Those come apart in a specific and predictable way. Reconstruction error is an average over dimensions, weighted by how much each contributes to that average, so a dimension is worth coding in proportion to its variance and to the variance it can explain in others. A dimension that is low-variance and weakly covariant with the rest is cheap to abandon: dropping it costs almost nothing on the objective, and the bits it would have consumed buy real reductions elsewhere. Under a tight enough budget the objective does not merely deprioritize such a dimension — it excludes it, and does so the same way across initializations, because the exclusion follows from the statistics rather than from where optimization happens to land.

That profile is a description of microstructure. Trade-flow imbalance and its companions are small relative to price movement, and their relationship to contemporaneous returns is weak enough that a code can reconstruct the bar without them. A microstructure-aware forecaster therefore depends on exactly the channels a reconstruction-trained tokenizer has the least reason to keep. The failure is not that the tokenizer is bad; it is that it is good at the objective it was given.

The difficulty in establishing this on market data is that a negative result there has too many explanations — no signal, too little capacity, too short a schedule. We therefore measure it on a constructed fixture where the answer is checkable: every non-price dimension carries identical marginal variance, and they differ only in dependence structure, so any difference in what survives is attributable to that one property. We then plant a signal of exactly computable information content and ask whether an end-to-end pipeline recovers it. It does not. The step that turns that into a diagnosis rather than another ambiguous negative is a control: the same backbone, at the same budget, recovers a signal planted directly in *token* space at 94.4% while recovering none of a larger signal planted in *feature* space. Model capacity is acquitted by construction, and the loss is localized to the interface between the two stages.

The remedy follows from the diagnosis and is an interface property rather than a matter of scale. The original design produces both halves of each token from one contextual encoder — causal, but free to look back — so bar t's state can be written into the codes of every *later* bar in the window, and a window-level decoder reconstructs it from all of them. The window reconstructs; the per-bar symbol carries almost nothing. Making the fine half a function of the current bar alone leaves per-bar legibility at chance — 0.4995 to 0.5182 across three seeds, against a baseline of 0.5112 — so the interface change is not sufficient on its own. Adding an objective that weights the microstructure block as a class, with a weight fitted on that same fixture, carries it to a seed mean of 0.9047 at unchanged bits per token and recovers the signal that was previously lost. Which half of that pair did the work matters later, and we keep them separate from here on. Because a fixture result need not transfer, that property is then converted into a gate that must pass on real data, on every one of the six live microstructure dimensions, before any second-stage expenditure is authorized.

Whether real trade-flow imbalance survives this pipeline and reaches a cost-aware trading decision is a separate question, and it is the one we set out to answer: a pre-registered, capacity-matched five-cell ablation with a shuffled-microstructure placebo, on 304,625,181 one-minute bars across 200 instruments, with the decision rule, thresholds, horizon, cost model, seed count and trial enumeration all fixed before any of their inputs existed.

It did not run — and what stopped it is the result. The gate fired, and it fired *selectively*: 97.3% of the shortfall falls on the two signed channels, trade-flow imbalance and signed count imbalance, while the four magnitude channels — the ones that co-vary with volume — essentially clear it. Under a placebo that shuffles microstructure against price the signed channels collapse toward chance and the magnitude channels hold. **The tokenizer keeps the microstructure that duplicates what OHLCV already carries and drops the part that is independent of it, which is the part a microstructure model would want.** We did not anticipate that split, and it is the paper's real-data finding. The gate described above fired on real data and stopped the study before any cell was trained through the second stage — an outcome the pre-registration had named as the mechanism's own modal prediction, and had bound in advance to make the mechanism finding the primary result. Section 6 reports what the gate measured and what was therefore never reached; this paper contains no ablation outcome — no cost-aware net information ratio, no null and no verdict. Figure 1 is that sequence in one panel. Three ungated baseline units were scored before the gate refused the microstructure arms, and what they support is confined to Section 7.

Two things this paper is not. It is not a foundation model: the backbone is a fixed measurement instrument, held identical across every arm so that the quantizer and the input arm are the only varied factors, trained on one venue at one frequency, with no transfer claim of any kind. And it is not a trading result — the cost-aware information ratio is the yardstick the ablation is scored on, not a strategy we are advancing. The claim is about tokenizers.

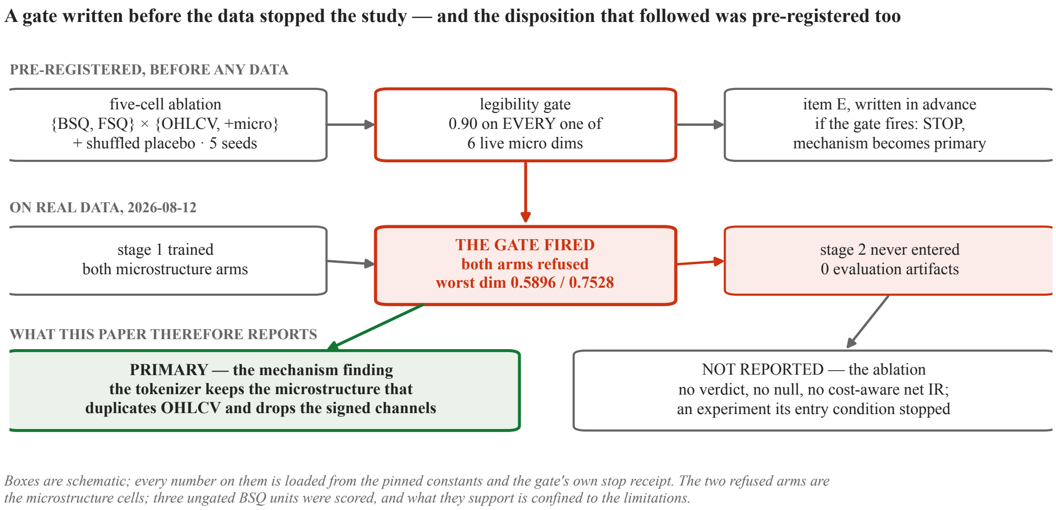

Figure 1: A gate written before the data stopped the study, and the disposition that followed was pre-registered too. The five-cell design, the legibility gate that stands in front of second-stage training, its firing on real data, and the pre-committed consequence — the mechanism finding becomes primary and the ablation reports nothing. The reading was not composed after the outcome: the rule that makes the mechanism primary on a firing was fixed thirteen days before it fired. Boxes are schematic; every number on them is loaded at render time from the pinned constants and the gate's own stop receipt.

Contributions.

- We identify and mechanistically characterize *capacity eviction*: under a reconstruction objective, code capacity is allocated by variance and covariance and never by downstream value, so a task-relevant channel that is low-variance and weakly covariant is excluded deterministically. We isolate the effect with a fixture in which the evicted dimension and the retained ones differ in exactly one property — statistical independence — and show the exclusion reproduces across seeds (Section 3).
- We localize the failure by construction rather than by inference: the same backbone at the same budget recovers 94.4% of a signal planted in token space and none of a larger signal planted in feature space, which acquits model capacity and indicts the tokenizer–backbone interface (Section 3.4).
- We introduce a per-bar tokenizer interface — a pointwise fine subtoken with a bottleneck decode leg and calibrated class weighting — that restores per-bar legibility at unchanged bits per token, and a *legibility gate* that enforces the property on real data as a pre-second-stage hard stop (Sections 3.6 and 3.7).
- We pre-register a capacity-matched five-cell ablation with a shuffled microstructure placebo over 304,625,181 one-minute bars, to be scored by cost-aware net information ratio under purged walk-forward validation, with the decision rule, seed count and multiple-testing correction fixed before any result existed — and then **the gate refused both microstructure arms before second-stage training, so that ablation was never run and this paper reports no ablation result** (Sections 4 and 6).

2 Related work

2.1 Foundation models for time series, and for K-lines

A line of recent work trains a single sequence model on large, heterogeneous collections of time series and evaluates it zero-shot or with light adaptation. Ansari et al. [2024] discretize scaled real values into a fixed token vocabulary and train a language model over the resulting symbols; Das et al. [2023] train a decoder-only model over patched inputs; Woo et al. [2024] train across frequencies

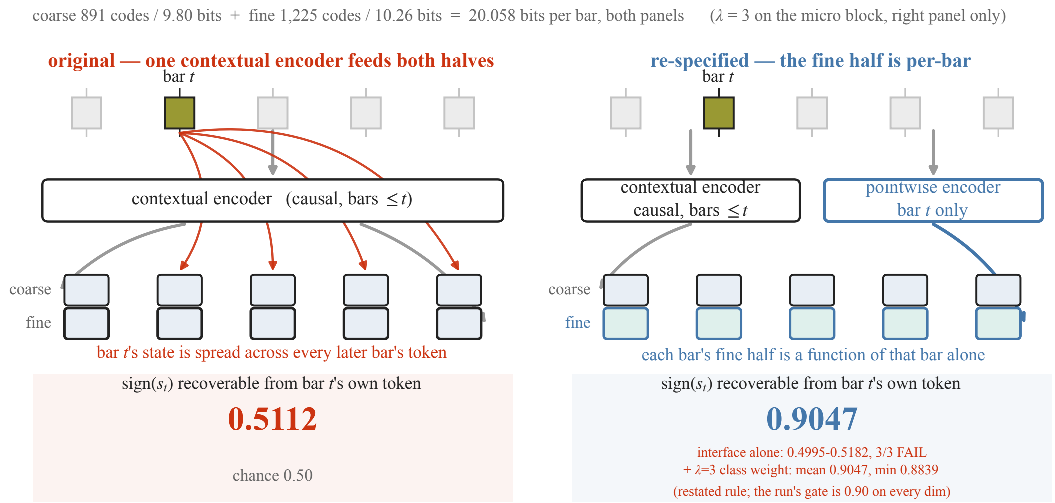

Figure 2: One interface change decides whether a microstructure signal survives tokenization. Each bar is encoded to a token pair — a coarse and a fine subtoken, together 20.058 bits — which is what the autoregressive stage consumes. **Left, the original design:** a single causal encoder produces both halves. Because that encoder is contextual, bar t's own state is distributed across the tokens of every later bar in the window, so the window reconstructs while the per-bar symbol carries almost nothing: the sign of bar t's microstructure state is recoverable from bar t's own token at 0.5112 on the fixture, against a chance rate of 0.50 (the token-control probe on the planted carrier reads 0.5135; both are chance). **Right, the re-specified design:** the fine half is produced by a pointwise encoder that sees only the current bar, shaped by a per-bar bottleneck objective that weights the microstructure block as a class. **The interface change alone does not restore recoverability** — 0.4995–0.5182 across three seeds, failing its own gate on all three — and 0.9047 is the seed mean reached only once the class weight is added, with a minimum of 0.8839 that clears a restated rule rather than the 0.90-on-every-dimension gate the real run applies. Bits per bar, the vocabulary split, the backbone and every evaluation instrument are identical between the two panels, so per-bar legibility is the only quantity that differs. What that difference buys downstream — the same planted signal recovered on the right and not on the left — is Figure 4; this panel is the interface, not the result.

and domains with a unified objective. Upstream of these, Nie et al. [2023] established the patch-as-token treatment that much of the line inherits. The shared premise is that a general representation of temporal structure transfers.

Shi et al. [2025] apply that premise to financial K-lines, and it is the design this paper starts from: a tokenizer trained to reconstruct per-bar market features, followed by an autoregressive model over the resulting discrete codes. We inherit the two-stage arrangement, the backbone dimensions and much of the training and evaluation scaffolding, and we vary one thing. Every component here is written from scratch; the parent's weights are not used, and no result in this paper is a comparison against them.

The artifact is a measurement vehicle, not a foundation model; Section 1 says what that means and Section 7.8 what it costs.

2.2 Discrete tokenization, and what its objective actually optimizes

Learned discrete representations descend from van den Oord et al. [2017], which pairs an encoder with a learned codebook and a commitment loss. Two later families remove the codebook. Mentzer et al. [2024] quantize each latent dimension to a small fixed set of levels, which eliminates the codebook, the commitment term and codebook collapse together, at the cost of a vocabulary that is a product of per-dimension level counts. Zhao et al. [2024] project to a hypersphere and binarize, giving an implicit vocabulary exponential in dimension. Yu et al. [2024] report that tokenizer design, rather than the generative model above it, is what determines downstream quality — a finding adjacent to ours in spirit and opposite in method, since it is established by improving a tokenizer rather than by measuring what one discards.

The relevant observation for this paper is about evaluation rather than architecture. These methods are compared on reconstruction fidelity, codebook utilization and downstream generation quality, and the first of those is the objective they are trained on. That is appropriate where reconstruction is the goal — an image tokenizer that reconstructs well has done its job. It becomes an assumption when the codes are handed to a forecaster, because the quantity that matters downstream is not how much of the input survives but *which parts* do. Reconstruction error is an average over dimensions weighted by their contribution to that average, so a dimension can be discarded almost entirely while the aggregate metric barely moves.

We are not aware of work that measures, for a reconstruction-trained tokenizer, which input dimensions are preferentially discarded and whether the discarded ones are the task-relevant ones. Section 3 is that measurement, on a fixture built so the answer is checkable, and Section 3.6 is an interface that changes it.

2.3 Microstructure as a predictive signal, and the distinction we are careful about

Order-flow imbalance is among the most robust short-horizon predictors in the microstructure literature. Cont et al. [2014] show that price changes over short intervals are driven largely by the imbalance between order-book events at the best quotes, a linear relationship with substantial explanatory power. Related work reads informational content from the composition of flow rather than its direction: Easley et al. [2012] construct a toxicity measure from volume-bucketed trade imbalance and relate it to liquidity provision.

The distinction that governs this paper is that these are not one quantity. Order-flow imbalance in the sense of Cont et al. [2014] is computed from *order-book depth* — the arrival, cancellation and execution of resting liquidity at the best quotes. What we compute is signed *executed-volume* imbalance from aggregated trades, which requires no depth feed and is available in the public dumps this study is built on. Trade signing itself has a literature [Lee and Ready, 1991]; our source labels the aggressor side directly, so no signing heuristic is applied and none of that literature's classification error enters our features.

We therefore use the name trade-flow imbalance throughout, in prose, configuration and code, and never the order-flow name. This is not pedantry about terminology. The published effect sizes belong to the depth-based quantity, they are not evidence about ours, and a negative result here is evidence about trade-flow imbalance only. Order-book depth is explicit future work (Section 7.10).

2.4 Evaluation discipline

The methodological literature this paper leans on is concerned with a single failure: that a sufficiently determined search over strategies will produce an impressive backtest from noise. Harvey et al. [2016] argue that the accumulated multiplicity of published cross-sectional predictors demands a substantially raised significance bar, and that most published findings would not survive one. Bailey and López de Prado [2014] give the deflated Sharpe ratio, which discounts an observed Sharpe by the number of trials it was selected from and by the non-normality of the return series. López de Prado [2018] supplies the sample-hygiene machinery — purging, embargoing and combinatorially purged cross-validation — that keeps overlapping labels from leaking a training window into its own evaluation.

Our design takes all four instruments: purged walk-forward with a measured embargo, a paired moving-block bootstrap, a deflated Sharpe ratio against an enumerated trial count, and a cost-aware net information ratio rather than a gross one. Section 4 specifies them.

What we add is not a new statistic but a procedural commitment. The deflated Sharpe ratio corrects for a trial count, and the trial count is supplied by the researcher; nothing in the instrument prevents an author from reporting a smaller one than they searched. Pre-registration is what makes that number checkable, and it is rare in this literature. We fixed the decision rule, the thresholds, the horizon, the cost, the seed count and the trial enumeration before any of their inputs existed, published the amendment log with the direction of each change recorded before the change was evaluated, and pre-committed a null as publishable. Section 8 describes the machinery; Section 8.8 reports what it caught.

2.5 Position

Against that background this paper makes one claim. Reconstruction-trained tokenizers allocate code capacity by variance and covariance and never by downstream value, so a task-relevant channel that is low-variance and weakly covariant is excluded — deterministically, under conditions Section 3.1 states — and microstructure is exactly such a channel. The remedy is an interface property rather than a bigger model or a longer schedule. Whether the same effect governs real trade-flow imbalance in a real market is a separate question, and it is the one Section 4 was pre-registered to answer.

3 Capacity eviction in reconstruction-trained tokenizers

Two-stage models for financial time series learn a tokenizer under a reconstruction objective and then fit an autoregressive model to the resulting discrete codes [Shi et al., 2025, van den Oord et al., 2017]. The design carries an assumption that is seldom stated: that *a representation good enough to reconstruct from is good enough to forecast from*. The tokenizer is fitted to reconstruction, the forecaster never sees the underlying series, and nothing in the pipeline is trained to preserve what the forecaster will need.

We show that the assumption fails, that it fails deterministically rather than by chance, and that it fails hardest on precisely the channels a microstructure-aware model depends on. We state the mechanism (Section 3.1), measure it (Section 3.2), test the prediction it makes about an end-to-end pipeline (Section 3.3), localize the loss by construction rather than by inference (Sections 3.4 and 3.5), give an intervention that removes it (Section 3.6), and convert the result into a standing gate that must pass on real data before any second-stage expenditure (Section 3.7). Every measurement in this section is made on a synthetic fixture with a known ground-truth signal, chosen so that the quantity being recovered is exactly computable; Section 3.8 states what that does and does not license.

3.1 The mechanism

A tokenizer maps each bar to a code of fixed size, so capacity is scarce and must be allocated across the bar's dimensions. The allocation is chosen by whatever the objective rewards, and a reconstruction objective rewards two things.

> **Claim.** Under a reconstruction objective, code capacity is allocated in order of the loss reduction each dimension buys. That ordering is set by a dimension's *variance*, which bounds how much loss it can contribute, and by its *covariance* with dimensions already coded, which determines how cheaply it can be served by capacity already spent. Downstream predictive value enters nowhere. A dimension that is low-variance *and* weakly covariant with the rest is therefore the last claim on the budget. Where two conditions hold together — a budget tight enough that not every dimension can be coded, and competing dimensions correlated enough that coding them is partly amortized — that dimension receives nothing, and does so deterministically: not occasionally, but in every run. Where either condition fails, the same objective still allocates by the same rule, but the exclusion is no longer determined; Section 3.2 reports a fixture in which the second condition is absent and the allocation is an initialization lottery instead.

The budget is tight, and the arithmetic says by how much. The canonical configuration uses per-dimension levels $(11, 9, 9, 7, 7, 5, 5)$, split into a coarse subtoken of $11 \times 9 \times 9 = 891$ codes (9.80 bits) and a fine subtoken of $7 \times 7 \times 5 \times 5 = 1225$ codes (10.26 bits), for 20.058 bits per bar; the binary-spherical arm [Zhao et al., 2024] spends $10 + 10$ bits for 20.000. Recovering the sign of a jointly-normal dimension at accuracy a requires a correlation $\rho = \cos\big(\pi(1 - a)\big)$, so the $a = 0.90$ we will require costs at least $-\frac{1}{2}\log_2(1 - \rho^2) = 1.69$ bits for that dimension alone by the Gaussian rate–distortion bound, and more under any realizable scalar quantizer. Thirteen live dimensions at that fidelity would need at least 22.0 bits — more than the entire per-bar budget, at any coarse/fine split. Not every dimension can be coded well. Something must be dropped, and the objective decides what.

This matters here because microstructure has exactly the profile the claim identifies. Trade-flow imbalance and the aggregate trade-flow statistics are low-variance relative to returns and only weakly

Table 1: Reconstruction quality by variance class, three seeds. Point-decoder correlation between a dimension's true value at bar t and its reconstruction from bar t's own fine code. Every dimension except the return has unit marginal variance; the signal dimension differs from the fillers only in being statistically independent. Filler entries give the range over the eleven filler dimensions; per-dimension values are in Appendix A.

dimension class	cross-dimensional structure	seed 0	seed 1	seed 2
return, sd 3.15	driven by the signal at lag 2	0.9818	0.9815	0.9823
fillers (11), sd 1.00	AR(1) chain across dims, ρ = 0.7993	0.826–0.918	0.824–0.919	0.824–0.917
signal, sd 1.00	independent	**0.0140**	**0.0011**	**0.0053**

covariant with price shape — they carry information about the near future precisely because they are *not* a restatement of the recent past. A tokenizer trained to reconstruct will therefore allocate capacity away from them by construction. This is not a defect of the objective; it is the objective working correctly against a goal it was never given. It also means that "microstructure-aware" has to be an architectural property of a tokenizer, and cannot be an emergent one.

3.2 Measuring the eviction

One point of order before the measurement, because this section is easy to read as describing the tokenizer introduced at the start of Section 3. The reconstruction results below were produced by a tokenizer that already carries interventions (i) and (ii) of Section 3.6 — the per-bar fine encoder and the bottleneck leg — which are not introduced until three pages from here. They are present because the eviction they expose survives them; the original contextual design fails the same test more severely, not less.

To measure the claim we need a fixture in which one dimension differs from the others in exactly the property the claim names, and in nothing else. We construct a synthetic bar stream in which the fixture is thirteen dimensions wide, matching the live width of the real vector, and stands to it as follows: one return dimension for price, eleven mutually dependent *fillers* standing for the OHLCV block and for the microstructure channels that co-vary with it, and one independent *signal* dimension standing for the channel whose predictive content is its own. The twelve that carry the comparison are marginally standard normal — so all twelve contribute equally to a variance-weighted loss — and then make eleven of them mutually dependent while one, the *signal* dimension, is independent of everything. A thirteenth, the *return* dimension, is deliberately *not* standard normal: it carries 3.15× the marginal standard deviation of the rest and stands in for price. Matching the twelve is what isolates dependence from variance; giving the return a larger scale is what supplies the high-variance channel the mechanism is about.

The dependence is not chosen for convenience; it is calibrated to the real data. On 30 symbols and 3,895,808 unmasked bars, the Gaussian-copula correlation log-determinant over the seven non-microstructure dimensions is −6.4819, a per-dimension entropy deficit of 0.4631 nats. The fixture reproduces that deficit with a cross-dimension AR(1) chain at $\rho = 0.7993$, achieving 0.462 nats against a 0.05-nat tolerance. One further dimension, the return, is driven by the signal dimension at a two-bar lag and so carries 3.15× the marginal standard deviation of the rest: it is the high-variance channel, and it is also the only dimension with downstream value in this fixture, since it is what a forecaster would predict.

Figure 3 and Table 1 report, for three independently initialized tokenizers, how well each dimension can be recovered from the bar's own code. The result sorts by variance class and does not vary across seeds. The return is recovered at 0.98. The eleven mutually dependent dimensions are recovered at 0.82–0.92, because one shared component serves all eleven at once and the capacity spent on it is amortized eleven ways. The signal dimension — same marginal variance as any single filler, differing only in being independent — is recovered at 0.001–0.014. It is not coded at all.

Calibrating the fixture to the real data is what makes this a mechanism rather than an anecdote, and we report the correction it forced because it changes the strength of the claim. On an earlier fixture whose dimensions were exchangeable i.i.d. draws — no cross-dimensional dependence at all — the allocation was arbitrary: across three initializations the set of dimensions clearing 0.9 changed completely, and the dimension of interest cleared in one seed of three. That reading makes

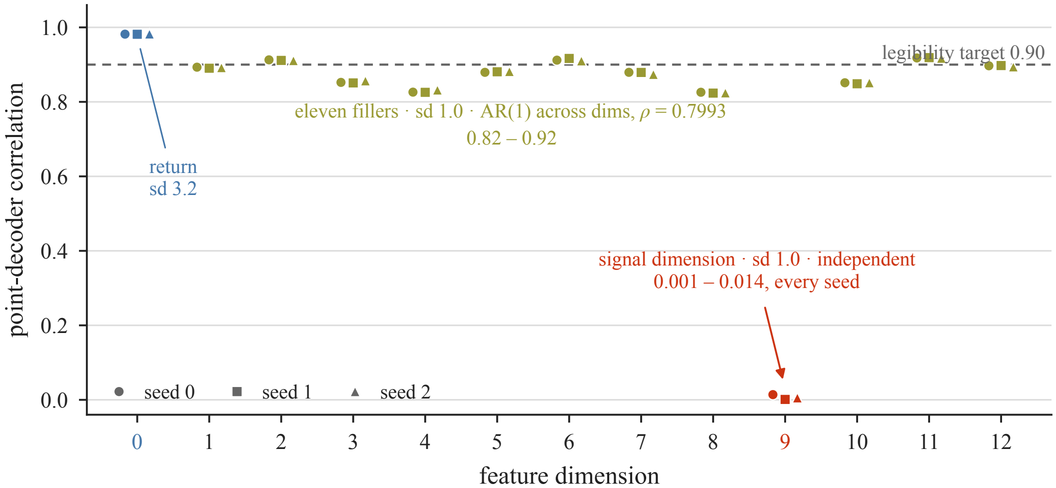

Figure 3: Per-bar recoverability sorts by variance class, not by task relevance, and the ordering is deterministic. Point-decoder correlation between each dimension's value at bar t and its reconstruction from bar t's own code, for three independently initialized tokenizers on a fixture whose non-signal structure is entropy-calibrated to the real data. The return dimension, whose marginal standard deviation is 3.15× that of the others, is recovered at 0.98; four of the thirteen dimensions clear the 0.90 rule in every seed. The eleven filler dimensions, which have unit marginal variance and are coupled by an AR(1) chain across dimensions at $\rho = 0.7993$, are recovered at 0.82–0.92. The signal dimension has *the same unit marginal variance as the fillers* and differs from them in exactly one property — it is statistically independent of everything — and is recovered at 0.001–0.014. The three seeds agree to within the marker width, so this is deterministic exclusion rather than an initialization lottery. Reconstruction objectives buy variance and covariance; they do not buy independence.

the phenomenon an initialization lottery, which is a substantially weaker and less actionable claim. Introducing the dependence structure the real data actually has replaced the lottery with deterministic exclusion, because the correlated block then offers an amortization the independent dimension cannot match.

3.3 The prediction, tested

If a tokenizer evicts a channel then a signal carried only by that channel is unavailable to everything downstream, however much information it contains and however capable the downstream model is. We test that prediction by planting a signal of known size.

We build a three-symbol stream of 42,000,000 bars. The signal dimension s_t is drawn i.i.d. standard normal and observed at the close of bar t; in the *planted* arm it drives the return two bars later,

$$r_t \;=\; \sigma\big(\varepsilon_t + c\, s_{t-2}\big), \qquad c = 3, \quad \varepsilon_t \sim \mathcal{N}(0,1) \text{ i.i.d.,} \tag{1}$$

and in the *noise* arm s_t is redrawn independently, so the arms differ in exactly one respect. Because the states are i.i.d., past returns reveal only states at lag ≥ 2, which are orthogonal to the forward label: the plant is recoverable *only* through the microstructure channel, and not through any function of price history. Equation (1) is a Gaussian channel, so the information it injects is exact,

$$I(s_t; r_{t+2}) \;=\; \tfrac{1}{2}\ln\Big(1 + c^2\Big) \;=\; \tfrac{1}{2}\ln 10 \;=\; 1.151 \text{ nats per bar,} \tag{2}$$

and because the tokens are a deterministic function of the features, the data-processing inequality makes 1.151 nats an upper bound on the validation cross-entropy improvement the autoregressive stage can obtain from the plant.

Both arms were trained under an identical budget and schedule — 20,000 steps at batch 32 and context 32, giving 20,480,000 bar-visits against a 27,300,000-bar training region, so 0.75 epochs, and memorization is not available as an explanation. The sub-epoch condition is asserted in the run receipt rather than assumed.

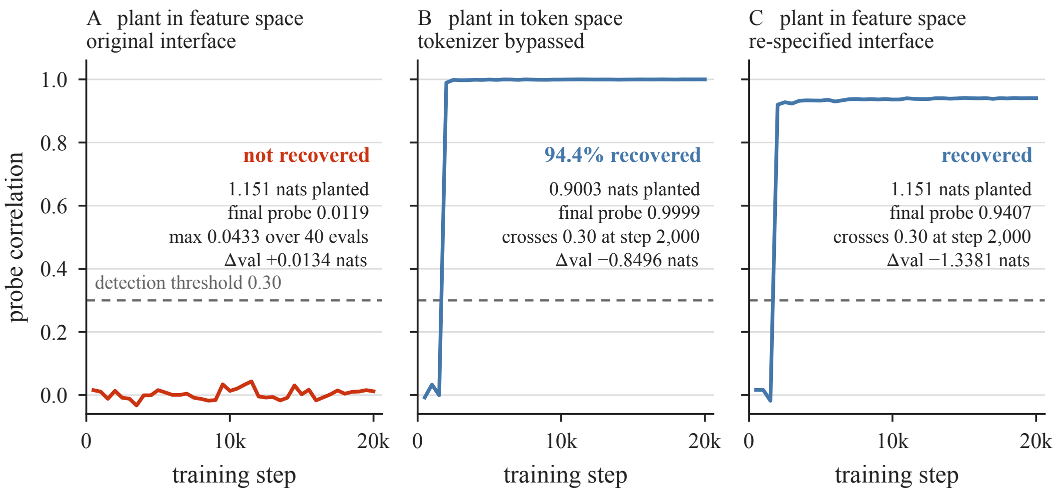

Figure 4: The same backbone extracts nearly everything from token space and nothing from feature space. *Plotted:* the forecast–state probe correlation against training step, with the dashed rule in each panel marking the pre-registered detection threshold of 0.30. The nats quoted below each panel are the planted and recovered information for the same runs, read from their receipts rather than from the curve. **Left:** 1.151 nats planted in feature space, where it must survive tokenization — the probe never leaves zero, staying below 0.05 at all 40 evaluation points. **Centre:** 0.9003 nats planted directly in token space, bypassing the tokenizer — the identical backbone at the identical budget recovers 0.8496 nats, 94.4% of what was planted, crossing the detection threshold at step 2,000. **Right:** the original feature-space plant, after the tokenizer interface is re-specified (Section 3.6) — recovery returns. Model capacity was never the constraint; the interface was.

The pipeline extracted none of it. The schedule completed on both arms with no loss spikes; the validation–train gap was +0.0072 (noise) and −0.0052 (planted), confirming the sub-epoch regime; the correlation between the model's forecast and the planted state stayed below 0.05 at *every one of the* 40 *evaluation points*, with a maximum of 0.0433; and the planted arm's final validation NLL was 13.6113 against the noise arm's 13.5979 — worse by 0.0134 nats, which is the wrong sign.

The tokenizer was not discarding the state. Window-level reconstruction of the planted dimension was non-degenerate in the same run: MAE 0.3044 against a predict-the-mean baseline of 0.8216. The information was present in the code stream and unavailable to the model consuming it.

3.4 Localizing the loss: a token-space control

A null admits two explanations: the tokenizer's code geometry hides the rule, or the backbone cannot learn thin conditionals at this scale. We separate them by construction rather than by argument, planting a signal of comparable size *directly in token space* — bypassing the tokenizer entirely — and asking whether the same backbone at the same budget recovers it.

The plant redraws a target digit from a discretized Gaussian centred monotonically on a source digit, with its width solved so the mutual information lands in a pre-registered band. It delivers 0.9003 nats exact over the full stream, agreeing with a Monte-Carlo plug-in estimate to 2.0×10^{-5}, with a marginal KL of 0.0048 and an entropy shift of −0.005 nats — so no information is smuggled in through the marginal — and an unplanted control returns -8.8×10^{-5}. Construction, calibration, and the retirement of a first, information-poor plant design are given in Appendix B. The carrier is the regenerated noise stream of Section 3.3 tokenized by that run's frozen tokenizer, so token distribution, budget, schedule and validation protocol are unchanged, and that run's noise arm supplies the reference $H_0 = 13.5979$.

The backbone recovers it almost completely: the probe crosses the pre-registered detection threshold of 0.30 at step 2,000 and reaches 0.9999, and the final validation NLL is 12.7483, that is $H_0 -$ 0.8496 nats, or 94.4% of the planted information. Figure 4 places the two experiments side by side against their planted-information references. The backbone is not the limiting factor. The loss is at the tokenizer–backbone interface.

3.5 The interface defect, measured directly

The window encoder is causal but *contextual*: bar t's code is a function of bars $\leq t$. State belonging to bar t is therefore free to be distributed across the codes of every subsequent bar in the window. The window as a whole reconstructs; no individual code need carry its own bar's features. Since the autoregressive stage consumes exactly the per-bar code sequence, anything that lives only in that spread-out form is unreachable.

The measurement is unambiguous. On 300,000 bars of the planted stream, tokenized by the frozen tokenizer of Section 3.3 with a 240,000/60,000 split, a logistic probe recovering $\mathrm{sign}(s_t)$ from bar t's own coarse and fine digits achieves 0.5135 against a chance rate of 0.5, and an MLP regressing s_t on the same digits achieves a Pearson correlation of 0.0142 — while the same tokenizer on the same stream reconstructs that dimension at the window level at MAE 0.3044 against a 0.8216 baseline. The state is encoded, and per-bar illegible.

A stronger form of the same defect appears when the encoder is not causal at all, and we record it because it is why causality is enforced. Perturbing only future bars inside a tokenization chunk and counting how many past bars' tokens flip gives a leakage rate of 41.75% coarse and 28.33% fine on a tokenizer trained on real data; the causal-encoder control measures exactly 0.0% on both, which is what establishes that the probe can read zero. Causal encoding is necessary for validity, and this section establishes that it is not sufficient for usability: it removes the leak forward and leaves the smear backward.

3.6 Intervention

If capacity is allocated by an objective that cannot see per-bar legibility, the remedy is to give the objective a term that can. We make three changes, and we state precisely what the evidence about them supports. They were introduced *cumulatively* — (i), then (i)+(ii), then (i)+(ii)+(iii) — so the ladder establishes that no *prefix* of the three suffices and that the full triple does. It does not establish that (i) or (ii) is necessary, because neither was removed from the final triple; only (iii) has that evidence, being inert at every λ without the others. **The supported claim is therefore that no prefix of the three was sufficient and the full triple was**, and we report the intermediate failures because they are what establishes it.

(i) A pointwise fine encoder. The fine subtoken becomes a per-bar encoding of bar t's own features, produced by a branch with no cross-bar mixing, while the coarse subtoken retains the contextual causal encoder. Both properties are carried in one token pair; bits per bar, the vocabulary split and the cell design are untouched. Alone this changed nothing — legibility 0.5202, at chance. The optimizer routed the state through the contextual coarse smear and spent the fine bits elsewhere.

(ii) A per-bar bottleneck decode leg. A pointwise decode head must reconstruct bar t from bar t's fine code alone, with zero cross-bar access, and its loss is added to the training objective. This trains — the dimensions that win reach point-decoder correlation above 0.9 — but it does not select *which* dimensions win: legibility 0.5136 and 0.5037 at canonical width, and 0.5014, 0.4995, 0.5182 across three seeds on the entropy-calibrated fixture, a 3/3 failure against the 0.90 threshold. That failure is the measurement plotted in Figure 3: the leg works, and the objective still spends its budget on variance and covariance.

(iii) Class weighting, with objective separation. The bottleneck loss weights the six microstructure dimensions as a class by a factor λ, and the window reconstruction losses receive the fine channels detached, so the fine encoder is shaped exclusively by the per-bar objective. Detachment is load-bearing, and was measured to be: with the window loss attached the per-bar weighting is inert at every λ — per-dimension receipts are identical from $\lambda = 2$ to $\lambda = 10^6$ — because the window objective re-creates the smearing incentive as fast as the leg removes it.

λ is calibrated, not chosen. The search ran over two knobs: the class weight λ and β, the weight on the bottleneck leg's own reconstruction term. Across 26 trials no (λ, β) pair cleared 0.90 on all three seeds, which places the threshold inside the instrument's seed-noise band at the achievable ceiling, while confirming that the state is carried (signal-dimension point-decoder correlation 0.77–0.96, microstructure class 0.86–0.92, return preserved above 0.96). The acceptance rule was therefore restated over the three calibration seeds as mean ≥ 0.90 and every seed ≥ 0.85, and λ set to the smallest searched value clearing it: $\lambda = 2$ gives $(0.8365, 0.8594, 0.8592)$, mean 0.8517 — fail; $\lambda = 3$ gives $(0.9060, 0.9142, 0.9000)$, mean 0.9067, minimum 0.9000 — pass. Re-running through the pinned construction path reproduces mean 0.9047, minimum 0.8839, bit-identically to the direct construction at fixed thread count.

We state plainly that this is an acceptance threshold restated after seeing calibration results, and why it is not the failure that description usually names. The restatement was forced by a measured property of the instrument rather than chosen to admit a preferred value: 0.90 sits inside the calibration instrument's own seed-noise band of roughly ± 0.03 at the achievable ceiling, so a fixed single-seed threshold at exactly that value tests the noise as much as the architecture, and the restated form — a mean clause plus a per-seed floor — is the standard response to that situation rather than a loosening of it. It occurred entirely on the synthetic calibration fixture, at a point when the real-data gate had never been executed and no real microstructure had been tokenized. And it did not propagate: the standing gate that governs expenditure (Section 3.7) requires ≥ 0.90 on *every one* of the six microstructure dimensions, with no mean clause and no floor, and nothing downstream of the calibration was selected on the restated form. $\lambda = 3.0$ is pinned in the conformance surface, applies identically to both quantizers and all input arms — so it cannot differentially favour any cell — and leaves bits per bar, the cell matrix and every evaluation instrument unchanged.

3.7 Verification, and the gate that carries it to real data

The intervention is verified by repeating Section 3.3 exactly: generator, seeds, stream size, plant strength, lag, budget, schedule and validation protocol unchanged, the tokenizer architecture the only difference. The forecast–state correlation crosses the detection threshold at step 2,000 and reaches 0.9407, with a maximum of 0.9414. The planted arm's final validation NLL is 11.8478 against the noise arm's 13.1859, a difference of -1.3381 nats where Section 3.3 gave $+0.0134$; the schedule completes without spikes and both arms stay in the sub-epoch regime. Per-bar legibility under the new tokenizer is 0.8990 on the planted arm and 0.9084 on the noise arm, where Section 3.5 measured 0.5135. We report Δval as the pre-registered decision statistic and not as an extraction fraction: each arm trains its own tokenizer, so the arm difference is not calibrated against the 1.151-nat plant in the way Section 3.4's control is calibrated against its own H_0. Figure 5 places every stage of the intervention against the gate.

A mechanism established on a fixture is a hypothesis about the real data, so we convert it into a gate rather than an expectation. After first-stage training and *before any second-stage expenditure*, each of the six microstructure dimensions must be linearly recoverable from bar t's own token pair at logistic sign-accuracy ≥ 0.90 on the run's real training stream; failure raises an exception and halts the run. The probe is a logistic regression on the one-hot digit decomposition of the coarse and fine codes, trained for 300 Adam steps at fixed seed on an 80/20 split, deterministic given its inputs. Sign accuracy is well defined for the magnitude-shaped dimensions because every microstructure feature enters z-scored, so the sign is above-versus-below the causal rolling mean, and per-dimension base rates are recorded so that residual class imbalance is visible. The sample is stratified by symbol with the split blocked in time *within* each symbol, which preserves both properties that matter: full universe coverage, and no near-duplicate neighbours across the split. Two defects in this gate were found and closed before it was relied upon, both of the form "a check that cannot fail is not a check"; they are documented with their fixes in Appendix C.

The gate's first execution on real microstructure is a named checkpoint with a pre-written adjudication: if the real dimensions cannot clear the threshold under the calibrated architecture, the run stops and reports with per-dimension receipts. It is not a threshold that may be moved. We record the expectation before the fact. Real trade-flow imbalance has the low-variance, weakly-covariant profile that Figure 3 shows being evicted, and the correction was calibrated on a synthetic plant. **A gate failure on real data would be the real-data confirmation of the mechanism** — simultaneously a blocker for the ablation and the strongest available evidence for the claim of Section 3.1. Both outcomes are reportable, and neither would be a surprise.

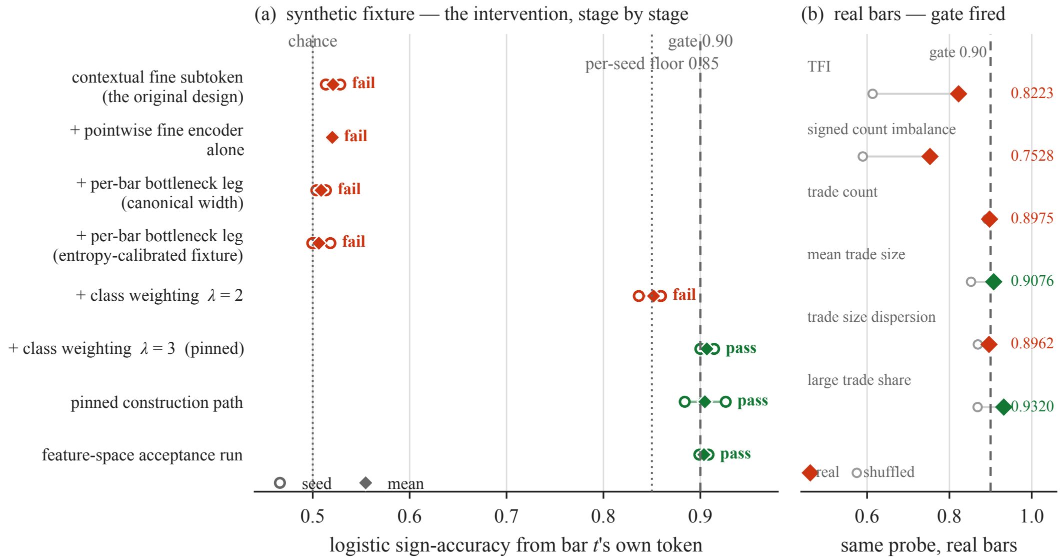

Figure 5: Only the third change moves per-bar legibility, and the threshold sits at the achievable ceiling. *Left:* logistic sign-accuracy of a microstructure dimension from bar t's own token, across the intervention. Open circles are seeds and the diamond is their mean; the acceptance rule over the three calibration seeds — *restated* after seeing calibration results, as Section 3.6 sets out — is mean $\geq$ 0.90 and every seed $\geq$ 0.85, so both thresholds are drawn. The original contextual design, the pointwise encoder alone, and the pointwise encoder with the bottleneck leg all sit at chance — the leg trains, but does not decide which dimensions win. Class weighting at $\lambda = 2$ still fails; $\lambda = 3$ clears, and the pinned construction path reproduces it. Note how little headroom exists above the gate: this is the achievable ceiling, not a comfortable margin. *Right:* the same probe applied to real microstructure by the standing gate, which ran after first-stage training and before any second-stage expenditure. **It fired.** Filled diamonds are the measured sign accuracies on real bars and open circles the same dimensions under the microstructure-shuffled placebo; the rule here is the gate's own, 0.90 on *every* live dimension with no mean clause, so both arms were refused. The shortfall is concentrated in the two signed channels, which are also the two that collapse under the shuffle while the four magnitude channels hold — the ordering the mechanism predicts, on market data (Section 6.1).

3.8 Scope of the claim

Four limits, restated in Section 7. First, everything above is measured on a *synthetic fixture*, not on real microstructure; the fixture's non-signal structure is entropy-matched to the real data and its signal dimension is a controlled construction, which is what makes the recovered quantity exactly computable and is also what stops it being a measurement of markets. The mechanism is therefore not this paper's headline claim, and is not elevated to one under any experimental outcome. Second, we demonstrate the effect for one reconstruction objective, one quantizer family and one class of encoder: we show that *this* objective allocates by variance and covariance, not that no reconstruction objective can do otherwise — and Section 3.6 is itself a counterexample to the strong form, being a reconstruction objective that carries the evicted channel because it was given an explicit term to do so. Third, the 94.4% of Section 3.4 is calibrated against a single matched reference run and inherits whatever variation that run carries. Fourth, λ is a hyperparameter fitted to a synthetic proxy for the real allocation problem; it is pinned pre-data and applied identically across every arm, so it cannot favour any cell, but the gate of Section 3.7 exists precisely because the proxy might not transfer.

4 Experimental design

The mechanism of Section 3 says that a reconstruction-trained tokenizer will evict microstructure unless it is built not to. This section describes the experiment that asks the separate question the paper is organized around: whether real trade-flow imbalance, carried through a tokenizer built to preserve it, survives to a cost-aware trading decision. The design was frozen before any of its numbers existed, and this section describes it as frozen.

4.1 Five cells

The experiment is a 2×2 over the two factors that could plausibly explain a difference — the quantizer and the input arm — plus a fifth cell that isolates the one thing a 2×2 cannot: whether an effect attributed to microstructure is really microstructure. Figure 6 shows the layout; the registry that the runner reads is the authority.

Cells 1 and 2 receive the seven OHLC-shape, volume and amount dimensions only; cells 3 and 4 additionally receive the six microstructure dimensions (indices 7–12: trade-flow imbalance, signed count imbalance, trade count, mean trade size, trade-size dispersion, large-trade share). Cells 1 and 3 quantize with binary spherical quantization; cells 2 and 4 with finite scalar quantization. Cell 5 is cell 4 with the microstructure block shuffled, and is the placebo.

The comparison that carries the paper is the paired difference $\Delta\mathrm{IR}(4-5)$. The others are support: $\mathrm{IR}(4) - \mathrm{IR}(2)$ asks whether the microstructure arm beats the OHLCV arm without reference to the placebo, and the two BSQ cells anchor the quantizer axis. We are explicit in Section 7 that the three marginals involving a BSQ arm are withdrawn as claims, because the BSQ implementation has no external reference point; they are reported, not claimed.

In the OHLCV-only arm the six microstructure dimensions are *removed* from the input rather than supplied and left uninformative, so that arm's feature width is seven rather than sixteen. That choice matters for interpreting the legibility gate, which is undefined for an arm with no microstructure channel and is exempted there by a named registry entry rather than by silence.

4.2 Capacity is controlled, not assumed

A comparison between two quantizers is meaningless if one of them simply has more room. We therefore control the two quantities that determine room, and report both rather than asserting parity. The finite-scalar arm uses per-dimension levels $(11, 9, 9, 7, 7, 5, 5)$, split into a coarse subtoken of 891 codes and a fine subtoken of 1225, for 20.058 bits per bar; the binary-spherical arm spends $10 + 10$ bits for 20.000. The two differ by 0.058 bits per bar, inside a pre-registered parity band of 19.5–20.5.

The predictor realizes 31,795,200 parameters at the finite-scalar vocabulary and 31,725,568 at the binary-spherical one. Of each, 10,493,952 — a third of the model — are the multi-token-prediction heads, which are trained and shipped; the backbone excluding them is 21,301,248 and 21,231,616 respectively, and that is the figure the construction gate pins. We quote both, and quote the total first, because the released weights make the total checkable in one line and a reader who computes it should find our number. The two arms differ by 69,632 parameters — 0.219% of the realized model, equivalently 0.327% of the backbone — arising entirely from vocabulary-dependent embedding and output-head rows. The heads are identical in both arms at 10,493,952 each, so the gap is the same absolute quantity on either basis. The backbone is otherwise identical across all five cells: eight layers, model width 512, feed-forward width 1024, eight heads, and a context of 512 bars. It is held fixed on purpose. The tokenizer is the object under study; the backbone is the instrument that measures it.

Every cell trains under the same budget: 26,003 steps at batch 32 and context 512, on the same draw. That budget is matched and fixed rather than chosen per cell by early stopping. The alternative — stopping each cell when its own validation loss saturates — would give five different budgets and confound $\Delta\mathrm{IR}(4-5)$ with "cell 4 trained longer", which is the same class of confound the placebo exists to remove. The cost is that no cell is guaranteed to be at its own optimum; saturation is measured and reported rather than optimized against.

4.3 The placebo, and the handicap it creates

The placebo answers a question no capacity control can: whether an effect attributed to microstructure comes from microstructure's *information* or merely from the model having six more input channels to spend capacity on. Cell 5 receives the same six channels, with the same marginal distributions, carrying no usable information.

The shuffle permutes the six microstructure dimensions as a block, within each contiguous segment of a symbol, with a full within-segment permutation rather than a fixed block length. Three properties are enforced rather than assumed. The block moves as a unit, so within-bar structure *among* the microstructure dimensions survives and only their relationship to everything else is destroyed. The mask travels with the value, because permuting values alone would strand fill values on valid bars and real values on masked ones — a defect found in review and fixed before any training. And the permutation is derived from the run seed and the symbol together, so the train and evaluation paths shuffle identically.

Figure 6 reports what the shuffle measurably does. It destroys the contemporaneous link between microstructure and price shape, from mean absolute correlation 0.2037 to 0.0005, and it destroys microstructure's own lag-1 autocorrelation, from 0.4165 to 0.0006. It leaves the within-block correlation at 0.2886 and the within-OHLCV correlation at 0.3105, both unchanged.

This is where the design's principal known weakness lives, and we state it rather than let a reader find it. Because cell 4's microstructure dimensions are correlated with price at 0.20 and autocorrelated at 0.42, part of the capacity they consume is *amortized* against structure the code is already carrying. Shuffling removes that amortization along with the information, so cell 5 pays a capacity price cell 4 does not. The consequence is a forfeit stated in the pre-registered rule string itself: $\Delta\mathrm{IR}(4-5)$ is (information) + (capacity handicap), and the paper does not claim the information effect alone clears any threshold. The handicap is not quantified in information-ratio units, and Section 7 carries that as an open disclosure rather than a resolved question.

4.4 The headline metric, the geometry, and the decision rule

The headline is a cost-aware net information ratio, annualized, computed on a portfolio return series. For a position of sign s opened at the close of bar $t+1$ and closed h bars later, the realized net log-return subtracts execution and funding cost directly:

$$r^{\text{net}} = s \log \frac{C_{t+h}}{C_{t+1}} - c_{\text{total}} - c_{\text{funding}}, \qquad c_{\text{total}} = 2\, c_{\text{side}}. \tag{3}$$

A position is taken only where the forecast magnitude exceeds a threshold proportional to cost, $\theta = \kappa c$, so the metric never rewards a forecast too small to pay for its own execution. Entry at $t+1$'s close, not t's, is the causal-safety margin: a decision made on bar t cannot transact at a price inside bar t. Information coefficient, rank information coefficient, mean absolute error and R^2 are reported as diagnostics and none of them decides anything.

Evaluation is purged walk-forward with an embargo, never a random split. The training region is used once and never refitted; five forward blocks carry the headline; a 120-bar embargo separates each training window from the labels it must not see, built as the 60-bar label look-forward plus a 60-bar serial-correlation margin. Section 7.9 reports the measurement that premise rests on, including the channel where it does not hold.

The decision rule was written, timestamped and committed before any cell was trained, and it is conjunctive: the primary outcome is SURVIVES only if none of five clauses fails, and NULL otherwise. The clauses require, in order, that the one-sided 95% lower bound of $\Delta\mathrm{IR}(4-5)$ exceeds zero; that the effect reaches the paired minimum detectable effect, so an effect too small for this design to resolve cannot be reported as one; that the lower bound of $\mathrm{IR}(4) - \mathrm{IR}(2)$ exceeds zero, a comparison not involving the placebo at all, so a placebo pathology cannot by itself produce a positive verdict; that $\Delta\mathrm{IR}(4-5)$ reaches an economic floor of 0.5 fixed before data, with the rule string naming the tested quantity as information plus capacity handicap so the manifest cannot be read as claiming the information effect alone; and that the deflated Sharpe ratio is at least 0.95 over the enumerated trial set. Two guards sit outside the clauses and are halt-only — they can refuse to report an outcome their inputs cannot support, and can never turn SURVIVES into NULL or the reverse. Appendix D gives the pinned constants, the exact clause strings, the trial enumeration, the minimum-detectable-effect derivation and the guards' asymmetries; they are machinery, and a reader does not need them to follow the result.

The design's own arithmetic said the likely outcome was negative, and we recorded that before running it. The minimum detectable effect at the pinned horizon is 3.518 annualized information ratio; our prior for a univariate microstructure signal at that horizon is a rank information coefficient of 0.0197. On these numbers, NULL or INCONCLUSIVE is the most likely honest outcome, and Section 7.4 states what that costs the design.

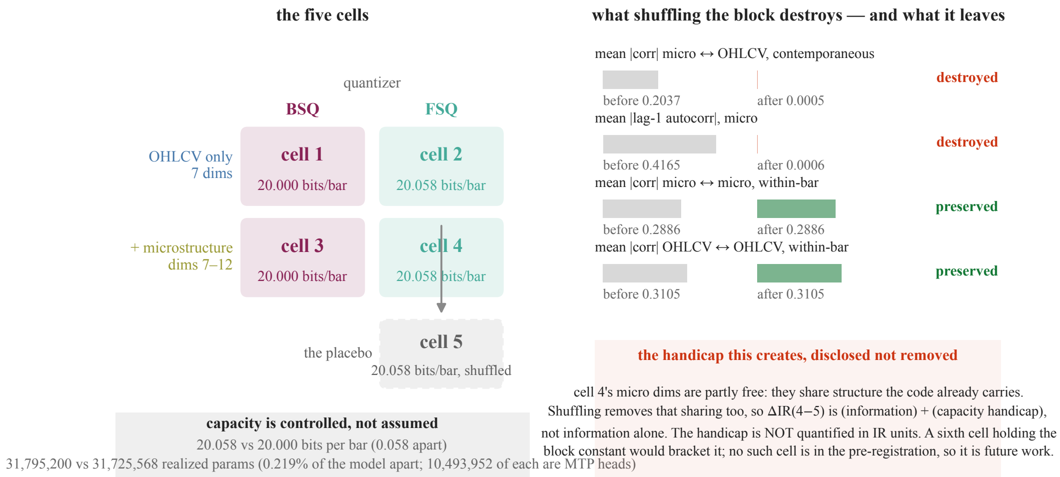

Figure 6: The five cells, at matched capacity, and the exact reach of the placebo. *Left:* a 2×2 over quantizer and input arm, plus a fifth cell that is cell 4 with the microstructure block shuffled. Capacity is controlled rather than assumed: the two quantizer arms differ by 0.058 bits per bar and by 0.219% of realized model parameters, so a difference between them cannot be attributed to one arm simply having more room. *Right:* what the shuffle actually does, measured. It destroys the two things it was built to destroy — microstructure's contemporaneous link to price ($0.2037 \rightarrow 0.0005$) and its own lag-1 autocorrelation ($0.4165 \rightarrow 0.0006$) — and leaves the within-block and within-OHLCV correlations exactly as they were. The consequence is stated rather than hidden: because cell 4's microstructure dimensions are partly amortized against structure the code already carries, shuffling removes that sharing too, so $\Delta\mathrm{IR}(4-5)$ is (information) + (capacity handicap). The handicap is not quantified in information-ratio units. A sixth cell holding the block constant would bracket it, but no such cell appears anywhere in the pre-registration, so it is future work rather than a committed contingency, and Section 7 carries the handicap as an open disclosure.

4.5 Pre-registration as the integrity mechanism

Everything above — the cells, the parity band, the placebo construction, the five clauses and their thresholds, the horizon, the cost, the κ grid, the seed count, the trial count and the null basis — is pinned on a conformance surface that the runner checks before it trains anything, and each pin carries a mutation test proving the gate rejects its predecessor value. Changing a pin is not a code edit that a reviewer must catch; it fails the run.

The pre-registration is a dated document with an append-only amendment log. Every amendment records what changed, when, why, and — where the direction was knowable — whether it made the bar easier or harder, recorded before the data that would reveal which. Amendments made after seeing calibration results are marked as such, including the one described in Section 3.6. The log also contains withdrawn claims: statements this project made and later retracted, retained rather than deleted. We regard that as the load-bearing part. A pre-registration that only ever records successful predictions is not evidence of discipline; one that records its own corrections is.

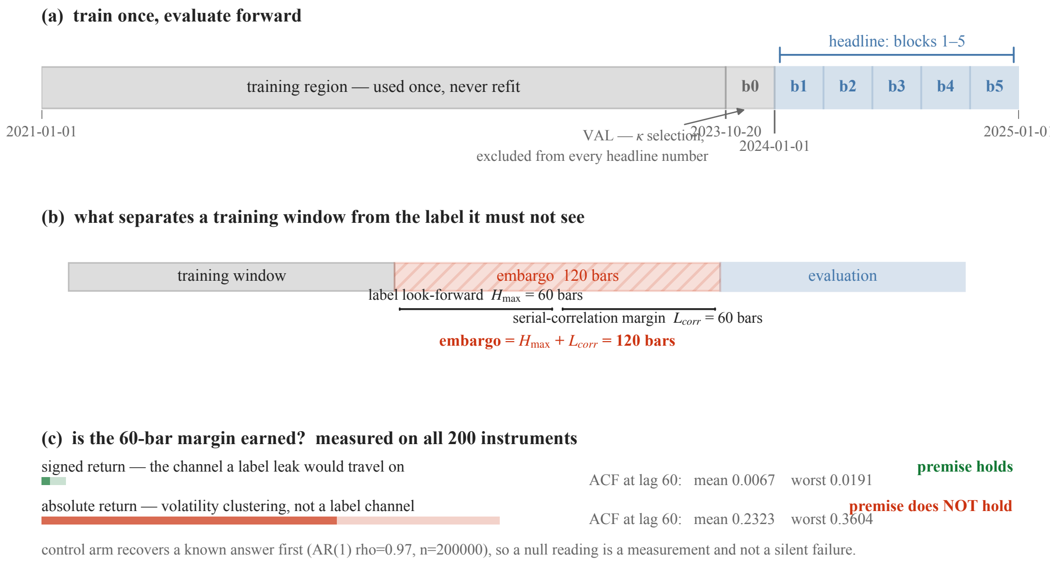

Figure 7: Train once, evaluate forward, and pay for the separation. **(a)** The training region ends at one boundary and is never refit. The forward region is cut into six blocks; block 0 is spent selecting the execution filter and is excluded from every headline number, so the headline is scored on five blocks the selection never saw. **(b)** Between a training window and any label that could contaminate it sit a 60-bar label look-forward and a 60-bar serial-correlation margin, giving a 120-bar embargo. **(c)** The premise behind that margin is measured on all 200 instruments rather than assumed, and a control arm recovers a known AR(1) answer first, so a null reading is a measurement and not a silent failure. It holds for signed returns — the channel a label leak would travel on — at ACF 0.0067 mean and 0.0191 worst-symbol at lag 60. It does *not* hold for absolute returns (0.2323 mean, 0.3604 worst), which is volatility clustering rather than a label channel; Section 7 carries that caveat.

5 Data

5.1 One venue, and the universe chosen from inside it

Every instrument is a USDT-margined perpetual future on Binance, sampled at one-minute frequency over the four years 2021-01-01 to 2025-01-01. Two public sources are combined per bar: the klines dump supplies open, high, low, close, volume and quote volume; the aggregated-trades dump supplies the signed trade statistics that the microstructure block is built from. No paid feed and no order book is involved, which is a scope limit (Section 7.10) and also the reason the corpus is reproducible by anyone.

The symbol set is bootstrapped *from the data* rather than from the exchange's live instrument list, and that choice is the one that decides whether the universe is survivorship-biased. A live instrument listing contains only instruments that still exist; selecting from it silently deletes every instrument that failed, which is the single most common way a backtest flatters itself. We instead scan the S3 index of the aggregated-trades dumps once, which yields per symbol, without downloading a single bar, the set of monthly dumps that exist. From that: the listing month is the first dump, and an instrument is treated as delisted when its last dump trails the newest global month. Ranking by total aggregated-trades bytes — a liquidity proxy available from the same index — and taking the top 200 gives the universe: **200 symbols, of which 41 were already delisted at the time of the scan**, each contributing only its [listing, delisting) bars.

One detail in that rule is worth stating because it changes three symbols. A dump index is published with lag, so an instrument that is merely a month behind the freshest global dump is not delisted — it is late. The delisting test therefore carries a one-month publish-lag tolerance, which corrected three symbols relative to a zero-tolerance comparison. Without it the universe would have carried three phantom delistings.

The realized ingest is **304,625,181** bars across all 200 symbols. That is below the 500M–1B band the blueprint nominated, and we flag it rather than quietly re-describe the band: 200 instruments over the fullest available window, many of them partial-coverage through late listing or delisting, lands near 300M. We regard it as adequate rather than merely acceptable, for a reason that is about the model and not about the data — a 21M-parameter backbone saturates well below this, so the lake is bought for regime and cross-sectional breadth rather than as training fuel. Reaching the nominal band would need roughly 320 instruments at proportional bandwidth, and would not train a better model.

5.2 What a bar is, and what three of its dimensions are not

Each bar is reduced to a 16-dimension feature vector. Seven dimensions are price shape and liquidity: log close-to-close return, log range, body fraction, upper and lower wick fractions, log volume and log quote volume. Six are the microstructure block, computed from aggregated trades: trade-flow imbalance $(v_{\text{buy}} - v_{\text{sell}})/(v_{\text{buy}} + v_{\text{sell}})$, signed count imbalance, log trade count, log mean trade size, trade-size dispersion, and large-trade share. These six — indices 7 through 12 — are the block the ablation varies, and the first of them is the quantity the paper is about.

The remaining three dimensions — funding rate, log open interest and its change, indices 13 to 15 — are specified, wired through every transform, and **structurally absent**. The v1 ingest covers klines and aggregated trades only, so on all 304,625,181 bars those three columns have standard deviation exactly 0.0 and their masks are set on 100% of rows. We state this here rather than in a footnote because the natural reading of "a 16-dimensional microstructure vector for perpetual futures" is that funding and open interest are in it, and they are not. The vector is 16 wide and 13 live. Section 7.10 carries what that costs.

Every feature carries a companion mask bit, and the mask travels with the value through every downstream transform. That pairing is not bookkeeping: a defect found in review had the placebo shuffle permuting values while leaving masks in place, which would have stranded fill values on valid bars and real values on masked ones. It was fixed before any training run.

5.3 Normalization that cannot see forward

Feature scales in this market move by orders of magnitude across four years, so the unbounded dimensions are normalized rather than used raw. The normalization is where lookahead usually enters a financial pipeline, through a statistic fitted on the whole sample and then applied to its own training region, so the rule here is absolute: every statistic used for bar t is computed from data available at the close of bar t.

Nine dimensions — the unbounded, scale-varying ones — receive a causal exponentially weighted z-score, at a half-life of 1,440 bars (one day) for the fast group and 5,760 bars (four days) for the slow volume and open-interest group. Seven dimensions are already bounded or relative, and are clipped only. Everything is clipped last to $[-5, +5]$ as an outlier guard. The volatility scale σ_t that the prediction targets are expressed in is built causally from raw returns within a segment, on the same rule.

Each statistic needs a warm-up before it is trustworthy, and warmed-up bars are dropped rather than back-filled. The cost is small and it is measured, not estimated: on the three thinnest instruments in the universe the warm-up drop is 809 of 279,930 bars, 811 and 809 respectively — about 0.3% — because most instruments are long single segments and the warm-up is a one-time tax at the start of each.

Causal safety is not asserted, it is a required continuous-integration gate. The test suite plants lookahead leaks of several kinds — a global one, a localized divisor leak, and a survivorship leak in the target-validity flag — and asserts that the exhaustive sweep catches each. One of those tests exists specifically to show that a *sampling* check misses the localized leak that the exhaustive check catches, so the gate cannot be quietly weakened to a sampler. During ingest the same sweep ran in-line on real cross-sections and a red result raised a fatal error before anything was recorded.

5.4 What the quality gates cost

Of the 304,625,181 bars, 303,283,972 (99.56%) **are usable by the OHLCV-only arm** after warm-up and quality masking, and 300,311,536 (98.58%) **are usable by the microstructure arms**. The ratio between those — microstructure availability given price availability — is 99.0% universe-wide, and no instrument falls below 0.7. Across the universe, 0.175% of bars are stale and 0.376% have zero volume.

That second number is the one that matters for the placebo. If microstructure were missing on a large fraction of bars, cell 4 and cell 5 would differ in coverage as well as in content, and the comparison the paper rests on would be confounded by absence rather than by information. At 99.0% availability with no starved instrument, it is not.

Forty-three instruments are thin, by a joint rule — fewer than half the median usable bars, or more than 10% stale. Inspecting them rather than dropping them: almost all are recent 2024 listings with 280k–500k bars and microstructure availability near 1.00, so they are thin by *age*, not by quality. The single genuinely illiquid instrument is a delisted one at 2.5% stale and 2.6% zero-volume. All 43 remain in the lake; thinness is handled as a sampling weight in the training draw, not as a deletion.

5.5 Coverage, and the survivorship we did not remove

Figure 8 is the whole universe as a symbol-by-month raster: 7,024 of a possible 9,600 symbol-months are present. The left staircase is instruments listing at different dates. The right edge is ragged because instruments that stop trading are *kept* in the lake, and their absence afterwards is real absence rather than a filter. A survivorship-cleaned universe renders as a solid rectangle, which is precisely the artefact the panel exists to demonstrate we did not create. Two instruments have interior gaps rather than truncated tails and appear as internal white streaks.

Two delisting counts appear in this project and they count different things, so we state both. The ingest record's **41** is the number of instruments known to be delisted as of the index scan, at any date. The **17** in Figure 8 is the number whose data ends *inside* the four-year window, which is all a raster over that window can show; the remaining 24 delist after 2024-12 and are complete here. The figure and this section therefore say "stop trading inside the window" and never "delisted".

Survivorship is owned, not solved. Retaining failed instruments removes one bias; it does not remove listing bias, since an instrument must have been listed on this venue at all to enter the universe.

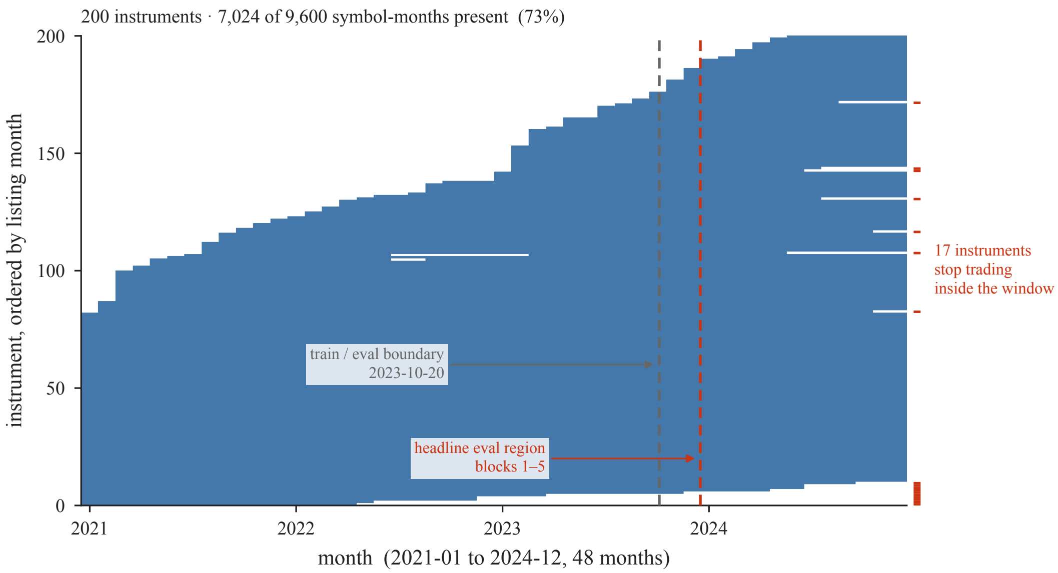

Figure 8: Survivorship is owned, not filtered away. Month-by-month presence for all 200 instruments over the four-year window, ordered by listing month: 7,024 of a possible 9,600 symbol-months are present. The left staircase is instruments listing at different times; the right edge is ragged because instruments that stop trading are *kept*, and 17 of them stop inside this window. A survivorship-filtered universe would render as a solid rectangle, which is the artefact this panel exists to show we did not create. Two instruments have interior gaps rather than truncated tails and appear as internal white streaks. The two dashed rules are the committed fold plan: training ends at the first, and the headline metric is scored only to the right of the second. Note that 17 counts instruments whose data ends inside the window; the ingest record's 41 counts every instrument known to be delisted at any date, including 24 that delist after this window closes.

Section 7.8 carries that.

5.6 Storage, provenance, and what makes a past prediction reconstructible

The reduced lake is Hive-partitioned Parquet under `(symbol, frequency, month)`, queried through DuckDB, one file per symbol-month. The layout was compacted for query performance, and the compaction was proven content-preserving rather than assumed: all 200 per-symbol dataset hashes were re-derived from the compacted lake and the universe Merkle root recomputed, with zero mismatches and an unchanged root.

Every dataset is content-hashed and the universe carries a single Merkle root over the lake, `sha256:5dfd667d05b97bda`..., recorded in the manifest alongside per-symbol listing and delisting dates and a corpus-purpose note.

Raw dumps are not retained, and that is a deliberate position rather than a shortcut. The Merkle root and the per-symbol hashes are taken over the *reduced* lake, which is the artifact we keep; the Binance SHA-256 of every raw shard is recorded for provenance only. Re-pulling raw and re-running a deterministic reducer reproduces the recorded lake hash — the reducer was proven byte-identical at the preceding milestone — so reproducibility survives eviction of 503 GB of raw input. Section 8 states exactly which parts of the pipeline are bit-exact and which are not.

5.7 What the experiment reads

The lake is broader than the experiment. Of the 200 instruments, the headline metric is scored on 40; the four-year window is split at its 0.7 point, on 2023-10-20, into a training region and a forward region, and the forward region is cut into six blocks of which block 0 is spent selecting the execution filter and excluded. The headline is therefore computed on blocks 1 through 5 — roughly 2024-01-01 to 2025-01-01, one year, 35,063 stride-15 periods per instrument — on a region no selection step has seen. The other 160 instruments and the preceding three years are lake breadth and training data, not evaluation breadth, and Section 7.8 states what that leaves unestablished.

6 Results

The pre-registered micro-legibility gate refused both gated arms before any second-stage training. Neither gated arm entered Stage 2 and neither produced an evaluation artifact, so the five-cell ablation has no result — not a null, and not an inconclusive one, but an experiment that its own pre-registered entry condition stopped. The gate reads the six live microstructure dimensions, so it binds only the microstructure arms; three ungated BSQ units were scored, and what they support is confined to Sections 7.5 to 7.7 and Section 8.1. Section 6.1 is therefore the whole of the measurement; Section 6.2 record what was not reached, and Section 6.3 carries the branch that pre-committed this disposition.

6.1 The pre-second-stage gate

The legibility gate of Section 3.7 runs on real data before any second-stage training is authorized, and it is a hard stop rather than a diagnostic: it requires at least 0.90 sign accuracy on *every one* of the six live microstructure dimensions, with no mean clause and no floor. **It fired.** Both gated arms were refused before any second-stage spend; neither entered Stage 2 and neither produced an evaluation artifact.

dim	feature	cell 4	cell 5 (shuffled)	base rate +
7	`TFI`	**0.8223**	**0.6135**	0.4869
8	`signed_count_imbalance`	**0.7528**	**0.5896**	0.4665
9	`trade_count`	0.8975	0.8916	0.4725
10	`mean_trade_size`	0.9076	0.8525	0.5462
11	`trade_size_dispersion`	0.8962	0.8691	0.4073
12	`large_trade_share`	0.9320	0.8685	0.8577
gate outcome		STOP	STOP	threshold 0.90

Two readings of that table matter, and they point the same way. **The shortfall is concentrated in the signed channels.** Measured against the 0.90 threshold, 97.3% of cell 4's total shortfall and 83.5% of cell 5's fall in dims 7 and 8 — trade-flow imbalance and signed count imbalance — while two of the four magnitude dimensions clear the threshold and two miss it by 0.0025 and 0.0038.

The placebo differential says it without depending on where the threshold sits. Against the majority-class baseline $\max(p, 1-p)$, cell 4 recovers the signed dims at lifts of +0.3092 and +0.2193; under the shuffle those collapse to +0.1004 and +0.0561, while the magnitude dims hold at +0.3641, +0.3063 and +0.2764. Destroying the microstructure block's own structure destroys the recoverability of the signed channels and leaves the magnitude channels standing.

One caveat belongs with the table rather than after it. `large_trade_share` carries a base rate of 0.8577, so its majority-class baseline is 0.8577 and its 0.9320 is a lift of only +0.0743 — the weakest recovery in the arm, on a dimension that nonetheless clears the threshold. The gate reads raw sign accuracy rather than lift, so an imbalanced dimension passes cheaply. This does not change the outcome, and it is recorded so that no reader takes that pass as evidence of recovery.

6.2 What was not reached

No cell was trained through Stage 2, so none of the five clauses of Section 4.4 was evaluated, no verdict manifest was emitted, and the primary outcome is neither SURVIVES nor NULL. The decision surface of Section 4.4 stands as specified and unexercised. The same holds for every secondary diagnostic — information coefficient, rank information coefficient, mean absolute error and R^2 — and for the dual-specification report, whose inputs do not exist.

The three marginals involving a BSQ arm, which Section 7.2 withdraws as claims in any case, are likewise unmeasured. Three BSQ units at the pinned budget — one cell, seeds 0, 2 and 4 — were evaluated before the gate refused the microstructure arms; they are not a cell comparison and no difference is computed from them.

The cost-stress curve and the seed-dispersion panel were pre-registered and are not drawn, because there is nothing to draw: no cell was scored. Their specifications stand in Appendix D and in the pre-registration, which is the right home for an analysis that has no data. The one quantity either would have carried that we *do* have — the measured break-even cost of the three ungated units — is reported at Section 7.6.

6.3 Interpretation

Five interpretation branches were written and dated before the run — four on 2026-08-10, one on 2026-08-11 — so that no reading could be composed with an outcome in front of us. The branch below is the one the outcome selected. The other four are deleted rather than softened, as pre-committed. This branch was additionally pre-written in the pre-registration itself — thirteen days earlier, in the final pre-data amendment, adopted 2026-07-30 against a firing on 2026-08-12 — as the modal prediction of the mechanism: "a firing is the MODAL prediction of our own finding … gate fires → STOP and report; the PRIMARY becomes the mechanism finding." It is now the primary.

The legibility gate fired on real data. The re-specified interface, which restores per-bar legibility on an entropy-calibrated fixture, does not restore it on real microstructure at this budget. What that establishes is the first non-fixture confirmation of capacity eviction: a reconstruction-trained tokenizer, built and explicitly weighted to carry the microstructure block, does not make it recoverable from each bar's own token.

And the failure is informative about where, which we did not anticipate. The per-dimension receipt localizes it: 97.3% of the shortfall falls on the two signed channels, trade-flow imbalance and signed count imbalance, while two of the four magnitude channels clear the gate and two miss it by 0.0025 and 0.0038. Under the placebo shuffle the signed channels collapse toward their base rates while the magnitude channels hold. **The tokenizer preserves the microstructure that duplicates what OHLCV already carries and loses the microstructure that is independent of it.** That pattern is what the mechanism of Section 3 predicts, and the prediction was made before the gate ran. **What is observed on real data is the consequence — which channels survive and which do not, with the evicted features named rather than inferred. That reconstruction is what allocates capacity is carried over from the fixture by argument, not established by an intervention on market data** (Section 7.1); Section 6.5 sets out how far the real-data evidence goes against a pure capacity explanation, and where it stops. It is also, precisely, the channel that invariant 1 scopes this study's claim to.

The claim ends there, and four limits bind it. It does *not* establish that no tokenizer can: λ was fitted on a synthetic proxy and, as Section 7.11 now records, on a single dimension that is itself a magnitude channel rather than a signed one. The budget is one point in a space we did not search. The threshold is a pinned constant whose calibration receipt already recorded that no searched configuration cleared it on all seeds. And it does not test the economic hypothesis at all — no cell was scored, so whether real trade-flow imbalance converts to cost-aware net information ratio remains unmeasured, and this paper does not answer it.

The pre-registered response executes without discretion: stop and report, no threshold moved, no rescope, no re-run at a higher weight. The pre-registration permits λ to be re-derived at most once, by the pinned formula, on a slice carved from the end of the training region, and binds the consequence in advance — the re-derived weight counts as an additional configuration in the multiple-testing correction, and any ablation conducted under it is reported as secondary and exploratory, never as a primary result. **The primary result of this study is the mechanism finding.**

6.4 A secondary, exploratory weight sweep — and what it says about the primary

Everything in this subsection is ***secondary and exploratory*** *by pre-registration. The protocol permitted the class weight to be re-derived at most once if the gate fired, and bound the consequence in advance: even a weight that cleared the gate would have returned the ablation only as a secondary, exploratory result carrying a doubled multiple-testing correction. Nothing cleared. The primary result of this study remains the mechanism finding of Section 6.3.*

The once-only allowance was spent on a sweep of the microstructure class weight across $\lambda \in \{5, 8, 12\}$, three seeds and both gated arms — eighteen configurations, trained on a calibration slice carved from the end of the training region. **None cleared.** The criterion was the same absolute one the gate applies, 0.90 on all six dimensions on both arms across all three seeds, and it was not moved.

cell 4, seed mean	$\lambda = 5$	$\lambda = 8$	$\lambda = 12$
`signed_count_imbalance` (dim 8)	0.7140	0.7187	0.7219
`TFI` (dim 7)	0.8268	0.8465	0.8574
OHLCV reconstruction MAE	0.12814	0.12678	0.12530

Table note, which travels with these numbers wherever they are quoted: ***these configurations are not comparable to the gate-firing run.*** *They were trained with the last tenth of the training region held out; the run of Section 6.1 was not.* ***No difference between the two sets is a measured quantity.*** *The criterion above is absolute and requires no baseline, which is what makes the sweep readable at all.*

Three readings follow. Two of them raise an objection to the primary finding before they support it, and we take that objection first rather than after, because it is the strongest one available against this paper.

The knob does not reach the failing channel. Across a 2.4× increase in the class weight, the worst dimension moves by +0.0079 — leaving 0.1781 of the gap to the threshold still open at the top of the swept range. The failure is not "one weight was too low"; it is that the dominant failing channel is close to insensitive to the only control the objective offers.

The weight does not buy capacity from price — on either arm. As λ rises, OHLCV reconstruction on cell 4 does not degrade; it improves slightly, from 0.12814 to 0.12530. On the placebo arm it is unchanged end to end, 0.17442 to 0.17444, a shift of +0.00002 across the whole range (the midpoint dips to 0.17313, so this is flatness between the endpoints rather than a monotone line). Weighting the microstructure class harder, by a factor of 2.4, takes nothing from the price channels.

Where it goes instead depends on how much headroom the block has, which is what saturation predicts. On cell 4, whose microstructure block is intact and already near the ceiling this tokenizer can reach, the gains are paid for internally: trade-flow imbalance rises +0.0307 while three of the four magnitude dimensions give ground, by −0.0045, −0.0063 and −0.0012, and the fourth moves +0.0025. On cell 5, whose shuffled block starts far lower — trade-flow imbalance at 0.7387 against cell 4's 0.8268 — there is room, and *all six* dimensions rise, the two signed ones by +0.0405 and +0.0248. The internal trade-off appears exactly where there is no headroom and vanishes where there is. Neither arm takes anything from price.

6.5 Is this a capacity ceiling rather than an allocation?

The two readings above invite a serious objection, and it is the one we would raise. If the class weight barely moves trade-flow imbalance and costs the price channels nothing, then nothing is being *traded off* — and a story with no trade-off in it is a story about a **ceiling**, not about **allocation**. The bit arithmetic of Section 3.1 sharpens it further: six microstructure dimensions at 0.90 sign accuracy need on the order of ten bits, which is approximately the entire fine-subtoken budget. On that reading the real-data failure is over-determined by arithmetic alone and needs no allocation mechanism at all.

We think the objection is right about capacity and wrong that it is the whole story, and the evidence that separates them is already in Section 6.1. Three observations, in increasing order of force.

First, a ceiling predicts a uniform shortfall and ours is sharply selective. A budget constraint does not know which dimensions are signed. If thirteen dimensions cannot fit at 0.90, the deficit should be spread across them. Instead 97.3% of cell 4's total shortfall falls in two of six microstructure dimensions; two of the remaining four clear the threshold outright and two miss it by 0.0025 and 0.0038. That distribution is not what running out of room looks like.

Second — and this is the load-bearing step — the placebo degrades *selectively*, which a capacity penalty cannot do. We state first what the placebo is *not*: it is not capacity-neutral, and we measured that ourselves. Shuffled microstructure is incompressible, so cell 5 faces a harder compression problem at the same budget, and it shows — cell 5 reconstructs the *byte-identical* OHLCV targets 1.51× worse than cell 4, worse on all seven price dimensions, with per-dimension ratios from 1.24 to 1.87. Same budget, less effective capacity. Any argument resting on "capacity is held constant across the placebo" is therefore false, and a referee who reads our own disclosure would be right to say so.

The argument that survives rests on the shape of the degradation within the arm, not on equality between arms. A capacity penalty is indiscriminate: a harder problem at a fixed budget degrades what the tokenizer encodes roughly together. What we observe under the shuffle is a 2.6× separation between two disjoint groups. The two signed channels fall by 0.2088 and 0.1632, collapsing to within 0.10 and 0.06 of their majority-class base rates. The four magnitude channels fall by 0.0059, 0.0551, 0.0271 and 0.0635 — the smallest signed drop is 2.6 times the largest magnitude drop. **A uniform capacity penalty does not produce a selective within-arm split along exactly the signed/unsigned line.** What the survivors have and the evicted channels lack is covariance with the price channels the objective already reconstructs.

Third, the channel is absent rather than merely hard to read per bar. On the fixture, the failure mode was that state was encoded but smeared across the window, so a per-bar probe missed what a windowed probe could find. That escape does not apply here. Reading trade-flow imbalance from the whole 512-bar window rather than from bar t's own token, with the same estimator and split and 19 log-spaced lags out to 511, moves sign accuracy from 0.8399 to 0.8385 — a change of -0.0014. The window recovers *nothing*. The signed channel is not hidden in the token stream; at this budget it is not in it.

The honest reading is that both are true, and stating both is stronger than choosing. Capacity sets the *size* of the loss: not all thirteen dimensions fit at this budget, and Section 3.1's arithmetic says so. Allocation determines its *address*: which two of the six were given up, and why those two. Neither claim does the other's work. What this study adds is the second — that the objective spends its bits on the microstructure that co-varies with price and abandons the microstructure that does not — and the placebo is what licenses it, because the placebo is the one comparison in the design where capacity is constant and structure is not.

The placebo turns out to be an eviction experiment in its own right. The shuffled arm's microstructure block is, by construction, independent of its own bar's price. Comparing the two arms dimension by dimension, the legibility gap is 0.0655 on the two signed dimensions against 0.0179 on the four magnitude dimensions — a factor of 3.66, averaged over all eighteen configurations. A channel stripped of covariance with its neighbours is systematically harder to encode. That is this paper's own mechanism, reproduced inside the control arm, on a manipulation performed for an entirely different purpose.

7 Limitations

The limitations below are stated at the same confidence as the contributions, and in the order a sceptical reader should apply them. Three of them — the synthetic basis of the mechanism, the absent external baseline, and the placebo's capacity confound — bound what the paper is entitled to claim even if every pre-registered clause passes.

7.1 The mechanism is measured on a synthetic fixture, not on markets

Everything in Section 3 is measured on a constructed fixture. The fixture's non-signal structure is entropy-matched to the real lake — a per-dimension deficit of 0.4631 nats, reproduced at 0.462 against a declared tolerance of 0.05 — and its signal dimension is planted with an exactly computable information content, which is what allows us to say that 94.4% of a known quantity was recovered rather than that a loss curve moved. It is also precisely what stops the result being a measurement of markets: we do not know the variance and covariance geometry of real trade-flow imbalance against real price well enough to assert that the fixture reproduces it; we know only that one entropy statistic matches.

The distinction that matters is between the mechanism and its consequence, not between the fixture and the paper's claim. What the fixture establishes is the *causal* statement — that a reconstruction objective allocates capacity by variance and covariance, shown by intervening on an entropy-calibrated stream where the answer is known in advance. What the gate establishes on real data is the *consequence*: this tokenizer does not make real trade-flow imbalance legible from a bar's own token, and under the placebo shuffle the signed channels collapse toward their base rates while the magnitude channels hold (Section 6.1). The second is the paper's headline claim and it is measured on markets; the first is its explanation and it is measured on a fixture. Neither substitutes for the other.

What a reader should therefore not take from the gate firing is the causal direction. The gate is an observation about an outcome: the signed channels are not recoverable. That they are not recoverable *because* reconstruction priced them out is carried over from the fixture by argument. The one real-data intervention we do have is the λ sweep of Section 6.4, which varies the objective's weighting but not its form, and Section 6.5 sets out exactly how far it and the placebo carry the causal claim and where they stop. The downstream ablation that would have separated the remaining alternatives was gated by the legibility check and never ran, so its evidence is absent rather than negative.

Two narrower limits inside the same fixture. We demonstrate the effect for one reconstruction objective, one quantizer family and one class of encoder; we do not show that no reconstruction objective can do otherwise, and Section 3.6 is a counterexample to the strong form. And the 94.4% recovery figure is calibrated against a single matched reference run, so we report it as one measurement rather than an expectation over runs.

7.2 The BSQ baseline is ours, and the external anchor was dropped

Cell 1 — binary spherical quantization on OHLCV-only inputs — is our own implementation at matched bits per token, not an independently validated reference. The harness was specified to reproduce a published reference model's rank information coefficient as an external anchor; that gate was found unexecutable and dropped as binding before any cell was scored, for three independent reasons. The reference paper publishes two rank information coefficients for the same model differing by 2.45× (0.0254 against 0.0622) without saying which applies; both are measured on Shanghai Stock Exchange 15-minute equity bars, a market and frequency this study firewalls out; and executing the weights would require importing reference model code that our independence invariant forbids. **We therefore cannot exclude that our BSQ baseline is weaker than a reference implementation, which would inflate any FSQ-versus-BSQ comparison.** That disclosure is a required field in every emitted verdict manifest, and the three BSQ marginals the design would have supported are withdrawn as claims rather than reported weakly. Resolvability was never the obstacle: the band is 4.51 to 11.05 standard errors wide at the measured effective sample size.

7.3 The placebo removes information and capacity together

Section 4.3 states the confound and quantifies what it can: the shuffle destroys the microstructure block's covariance with price and its own serial structure, and it also makes the block incompressible, so cell 5 reconstructs the byte-identical OHLCV targets 1.51× worse than cell 4. **The residual difference between cells 4 and 5 is therefore information plus capacity, and we cannot separate them by subtraction.** No cell in the design brackets the capacity term, so it is disclosed and unquantified in information-ratio units; a sixth cell holding the block constant would bracket it and is future work. Section 6.5 gives the one reading that survives the confound — the shape of the degradation within the arm rather than any comparison between arms.

7.4 The effect we can detect is larger than the effect we expect

The minimum detectable effect at the pinned horizon is 3.518 annualized information ratio, and our own prior for a univariate microstructure signal is a rank information coefficient of 0.0197 at that horizon. **The design was therefore powered to detect an effect substantially larger than the one we had reason to expect**, which is why Section 4.4 pre-committed that NULL or INCONCLUSIVE was the most likely honest outcome. Two facts sharpen it. The three-seed replication carries no term for training variance, and same-seed divergence measured on identical hardware puts a lower bound on that term; the power guard exists for exactly this and is halt-only. And the effect the design can see is one the execution filter may never admit — Section 7.5.

7.5 The execution filter binds on real data, and the forecasts are over-dispersed

The headline metric is a cost-aware net information ratio under a forecast-magnitude filter, $\theta = \kappa c$. On the three ungated units that were scored, decision activity at κ^* is 0.0926, 0.0193 and 0.4702: the filter is neither inert nor absorbing. **What that reveals is a calibration failure, not a healthy**

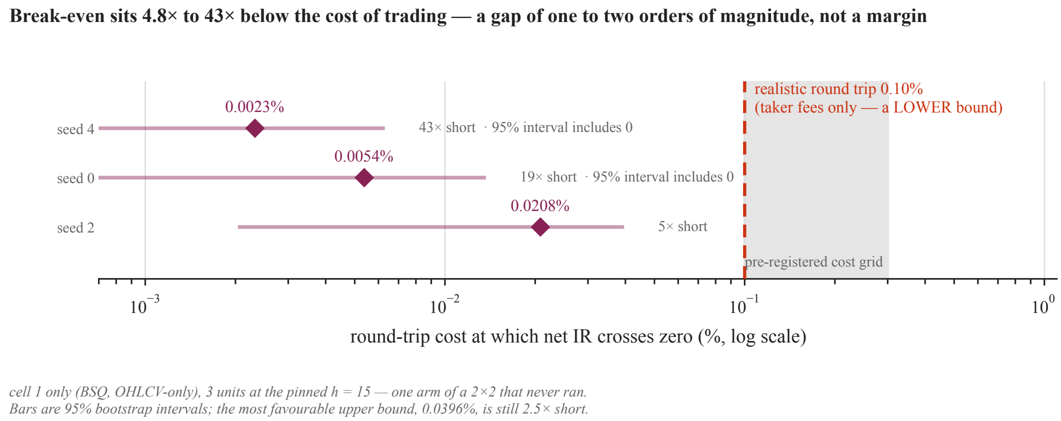

Figure 9: Break-even sits one to two orders of magnitude below the cost of trading. The round-trip cost at which net information ratio crosses zero, for the three scored units, against a 0.10% taker round trip. The axis is logarithmic because the finding is the *size* of the gap: on a linear axis every measured value collapses onto the origin. The shaded band is the pre-registered cost grid, whose floor is the comparison itself — which is why the pre-registered break-even statistic returns $-\infty$ here and cannot resolve the question. Intervals are 95% bootstrap; two of the three include zero. Cell 1 only.

book. A calibrated forecast at our own prior would have $\mathrm{std}(\hat{\mu}) = 0.027 \times 0.003504 = 9.46 \times 10^{-5}$; the measured values are 0.005376, 0.005331 and 0.010373 — 56 to 110 times that. The models are grossly over-dispersed, which is why they trade at all, and it means a realized information ratio would reflect miscalibration at least as much as skill. We had reasoned from the pinned budget being compute-optimal to the models being calibrated and had planned for the opposite branch; the measurement selected the branch we deprioritized.

7.6 Break-even cost falls an order of magnitude short of fees

This is the number a reader asking "how far from tradeable is any of this?" should take away. At the pinned horizon the round-trip cost at which net information ratio crosses zero is 0.00232%, 0.00538% and 0.02082% across the three scored units, against a 0.10% taker round trip — 4.8 to 43 times short. It survives sampling error: the most favourable 95% upper bound is 0.0396%, still 2.5 times short. Two qualifications are ours to state. The 0.10% threshold is supplied rather than measured here, though the direction rescues it, since a taker round trip excludes spread, impact, funding and slippage and the true bar is therefore higher. And the interval statement is not independent corroboration of the gross-information-ratio statement: gross IR *is* the t-statistic of break-even on this grid.

Our own pre-registered instrument could not have told us this, and the defect is worth recording because it is structural rather than clerical. The cost-stress curve sweeps a pinned grid whose *floor* is 0.10% — exactly the comparison — and the break-even statistic returns $-\infty$ when net IR is non-positive across the whole grid, which it is on all three units. The instrument cannot distinguish a break-even just below a round trip from one forty-three times below it. A second defect compounds it: the value *is* computed at evaluation time and the artifact writer does not persist it, so it was computed and discarded.

The strongest form of the result is assumption-free. Break-even equals mean gross return per active period identically, so clearing a round trip of c requires gross edge per active period of at least c — and that bar is the same at every horizon. What a longer horizon changes is how much edge a period can contain, not the bar it must clear. Under linear scaling, which is generous because real drift decays, the measured edge would reach 0.10% only after 72 to 646 minutes of holding against the 15 evaluated. **This evidence is weak in both directions:** it covers cell 1 alone, so a horizon that cleared fees would have been suggestive rather than proof, and finding none does not license "no horizon is tradeable" — the arm the headline claim is about was never scored.

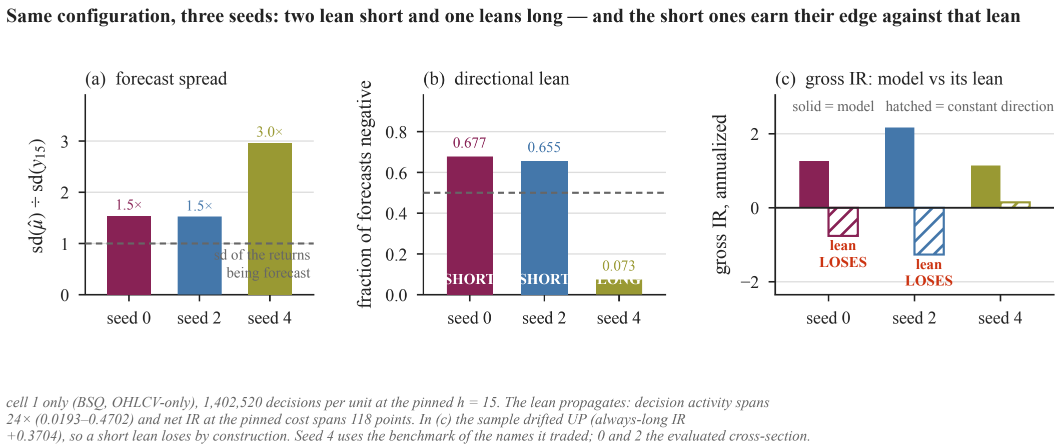

Figure 10: Three seeds of one configuration take structurally different directional positions — and the two short-leaning units earn their edge against that lean. *(a)* the forecasts are 1.5–3.0 times wider than the returns they forecast. *(b)* two units are negative on roughly two-thirds of decisions and one on seven per cent, a difference of kind rather than degree. *(c)* each unit's gross information ratio beside the same unit's own modal direction held constant: for seeds 0 and 2 the lean *loses* money in a sample that rose, so drift capture predicts the wrong sign for them. The lean propagates — decision activity spans 24× and net information ratio spans 118 points across three units that differ only by seed. Cell 1 only, 1,402,520 decisions each.

7.7 The positive gross edge is not directional bias

A positive gross edge that is never tested against the cheapest alternative explanation is an assertion, so we ran that test. The alternative is directional bias times sample drift: the evaluated sample rose at 7.900×10^{-6} per period, enough that simply holding long returns an annualized information ratio of 0.3704, and each model leans one way. The benchmark is each model's own modal direction held constant on the same active periods, with a paired block bootstrap.

On two of three units the bias explanation does not merely fail — it predicts the wrong sign. Seeds 0 and 2 lean short in a sample that rose, so their lean alone *loses* money, −0.7607 and −1.2640, while the models earn +1.2665 and +2.1715. On the third, measured against the names it actually traded, holding long reproduces 12.9% of its gross information ratio and the remainder is +0.9972. **That margin is thin and we quote every bound with its level.** The pre-registered instrument is a one-sided bound at $\alpha = 0.05$, where the percentile bound is +0.0454 and its normal counterpart +0.0307; a two-sided 95% bound is a stricter test and does not clear, at −0.1545 with $p = 0.0897$. The effect is real on the instrument declared in advance and it is not comfortably large. **None of this is a claim of economically useful skill** — Section 7.6 puts the gross edge an order of magnitude short of fees. It is the narrower claim that the cheapest deflationary explanation is refuted.

7.8 One exchange, one venue, and a scored universe of forty

The lake is 304,625,181 one-minute bars across 200 instruments, and that breadth is not breadth of evidence. Every instrument is a USDT-margined perpetual future on a single exchange; perpetuals are not spot, fee schedules and liquidation mechanics differ across venues, and trade-flow imbalance is venue-local in a way returns are not. **More narrowly still, the evaluation universe is forty instruments, not two hundred**: the gate's probe drew 150,000 decisions from forty, and the money grid scores forty. The two hundred is the training draw. One scored year cannot separate a regime-specific result from a general one, and the year is a single crypto cycle phase.

7.9 The embargo premise holds for one channel and fails for the other

The 120-bar embargo is built on a measured premise, and the measurement is honest about where it does not hold. Signed-return autocorrelation at lag 60 is 0.0067 on average and 0.0191 at worst, so the margin is comfortable for the channel a label leak would travel on. Absolute-return autocorrelation is 0.2323 and 0.3604 — volatility clustering, which is not a label channel but is not nothing either, and a volatility-conditioned strategy could in principle exploit it across the boundary. We state both rather than the favourable one.

7.10 "Microstructure" here is narrower than the word suggests

Three limits compound. The imbalance is computed from executed trades, so it is *trade*-flow imbalance and not order-flow imbalance; true OFI needs book depth, which v1 firewalls out. Funding and open interest occupy three of the sixteen feature dimensions and are masked on every bar in the lake — specified, wired, and structurally absent. And the class weight $\lambda = 3$ was fitted on the synthetic fixture and on a single decision path, then pinned; the gate is what guards the transfer, and on real data it refused.

7.11 Matched to a tolerance, not exactly

The two quantizer arms differ by 0.058 bits per bar and by 69,632 parameters — 0.219% of the realized model, 0.327% of the backbone alone; we give the denominator because a bare percentage travels without one. Both gaps favour the FSQ arm and both are an order of magnitude smaller than any effect the design can detect. Separately, the acceptance rule for the interface fix was *restated* after seeing calibration results, from a single threshold to a seed mean plus a per-seed floor. Section 3.6 sets out why that is defensible and what it costs; the run's own gate is the stricter absolute form, applied to every dimension with no mean clause.

7.12 Reproducibility is bit-exact except in training

The data pipeline, the frozen normalization statistics and prediction replay are bit-exact from one configuration file, a pinned seed and content-hashed inputs, within one provenance identity. GPU training is not, unless the deterministic-attention fallback is enabled, and deterministic attention is necessary but not sufficient: reduction order is unconstrained unless deterministic algorithms are forced as well. Section 8.1 reports what replay costs across environments and what it does not cover.

8 Reproducibility and verification

8.1 What is bit-exact, and what is not

Reproducibility here is a deliverable rather than hygiene, and it is scoped rather than asserted whole. Three things are bit-exact from one configuration file, a pinned seed and content-hashed inputs: the data pipeline, the frozen normalization statistics, and prediction replay. Any recorded prediction can be reconstructed from the recorded inputs.

That claim is scoped to one environment, and we can now say what happens outside it. Rescoring a banked predictor on a different `torch` build and a different physical GPU, weeks later, reproduces the scored series *closely but not bit-exactly*: the set of active periods is identical at 34,989, the gross series agrees to a maximum absolute difference of 2.59×10^{-4} with a correlation of 0.99999986, and the information ratio moves by 3.5×10^{-4}. A different build is a different provenance identity by construction — the `torch` version and the platform ABI are both identity keys — so this does not contradict the bit-exactness claim, which was never asserted across identities. It bounds what the claim does not cover, which had until now been untested: replay across environments is reproducible to about four decimal places, not to the bit.

Four limits belong with it, and we state them rather than let the agreement figure travel alone. It is *not* bit-exactness — the residual 2.59×10^{-4} is the float32 reduction-order effect that deterministic attention is already conceded not to remove. It says nothing about *training* reproducibility, which is where the same-seed divergence of Section 7.12 lives and which this does not touch. It covers one arm and one seed, so it is not a general claim about the pipeline. And both environments ran with deterministic attention and deterministic algorithms forced, so it establishes nothing about the default path. What it does establish is the half that had been argued rather than shown: the scored inference path survives a change of `torch` version and of physical machine, weeks apart, with a bit-identical set of active decisions.

GPU training is not bit-exact by default, and the honest version of that sentence took us a correction to write. The production path now sets deterministic algorithms explicitly, at a measured throughput cost of 13.175%, and every run records the mode it ran in.

The correction is worth stating because it is the kind of error a reproducibility section normally hides. For a period, 49 of 49 run records carried an assertion of bit-exactness beside a configuration that had determinism switched off. The cause was a single line that computed the claim as *attention mode equals the deterministic kernel* — our own invariant's sufficiency premise encoded as code. Deterministic attention is necessary for bit-exact training and is not sufficient, because reduction order elsewhere in the backward pass is unconstrained. The scope of the false claim was established at the time and is narrow: only the GPU-training clause was affected; the pipeline, frozen-statistics and prediction-replay clause held, the anchor of Section 8.2 held, and no prior result was invalidated. We report it here rather than quietly shipping the fix because a reader assessing our other claims should know that this one was wrong for a while and how it was caught.

One consequence survives the fix. Same-seed runs were observed to diverge — fraction-negative 0.5662 against 0.99883 for the same cell at identical horizon, cap, estimator, seed and recipe — which is basin-hopping, and a lower bound on across-seed variance. Forcing determinism removes the same-seed component and leaves the across-seed component untouched, so it does not repair the statistical consequence. Section 7.4 carries that; Section 8.7 describes what we do about it.

8.2 The anchor

A reproducibility claim that is only checked at the end is a claim about one moment. We instead carry a standing anchor: a fixed evaluation of a frozen checkpoint through the full harness, whose results hash is re-proven before and after any change to the evaluation instrument. The current anchor is `sha256:3f86882a63dd06c78…`, over inputs that are themselves content-hashed — dataset, tokenizer and predictor each carry their own SHA-256 — and it is accompanied by an exhaustive causal sweep at 420 of 420 checks.

Two procedural rules attach to it, both of which exist because of a specific failure. The anchor is re-proven *twice*, from separate invocations. And its exit status is read from an unpiped command, because an exit code read through a pipe is the pipe's — a '`| grep | tail`' once masked a non-zero exit and let a false anchor claim into a commit message. The anchor has been legitimately re-based once, when a file inside the anchored instrument changed for a correctness fix; the re-basing is recorded rather than presented as continuity.

The environment is pinned by a lockfile: Python 3.11.7, NumPy 2.4.6, PyTorch 2.12.1. That pin is itself a response to a defect — environment drift with no lockfile — and load-bearing results are re-verified under it rather than under whatever interpreter happened to be first on the path.

8.3 Content addressing

Every produced dataset is content-hashed, and the universe carries a single Merkle root over the lake, recorded in the manifest with per-symbol listing and delisting dates (Section 5.6). The property this buys is specific: any past prediction can be reconstructed, because the exact bytes it was computed from are identified rather than described. The compaction of the lake was proven content-preserving by re-deriving all 200 per-symbol hashes and recomputing the root, rather than by inspecting the compactor.

8.4 Pre-registration, and an amendment log that records the direction

The decision surface — clauses, thresholds, horizon, cost, the κ grid, the seed count and the trial count — was fixed before any of its inputs existed, and the design was frozen before the run. Amendments are not forbidden; unlogged amendments are. Every change is recorded with its date, its reason and, crucially, a *direction-blind* test: before looking at whether the amendment helps, we record whether it tightens or loosens the criterion. The log carries 58 distinct revision tags.

One of those amendments was made after the run began, and it is flagged as such wherever it bears. Every other entry predates the existence of any evaluation unit. The exception rescoped a codebook-utilization gate to the quantizer the specification had scoped it to, after two units had been evaluated. We hold it to be a restoration rather than a relaxation — the specification sets that target for one quantizer, calls the diagnostic a health signal rather than a failure mode, and says the other quantizer's figure is *reported*; the gate had been transcribed into our own log without its scope.

But a reader is entitled to the mechanical fact independently of our reading, so the log records it as a loosening, records what was visible when it was made, and records the counterfactual: had the measurement come out the other way, the same restoration would still have been correct. One consequence is pre-committed rather than left to discretion — the two units evaluated at that point bear on the direction of the secondary fallback comparison, so **any fallback claim arising from this run is reported as evaluated under a post-data amendment**, whether or not the fallback fires.

The direction test is not decorative, and it has already overturned our own summary of our own amendments. The most consequential window changed the multiple-testing trial count and the dispersion basis for the deflation benchmark — moved from all arms to the placebo arm alone — together, and it was written up as *every amendment in this window loosens the effective bar*. That was false. The deflation clause does loosen, by a factor of 0.859. But the clause that most directly tests the headline contrast *tightens*, twice over: its quantile multiplier rises from 2.4865 to 3.0728 at five seeds, a factor of 1.236, and the total standard error gains a training-variance term that can only increase it. The net across a conjunctive gate is indeterminate before the run and is not monotonically loosening. Had the original framing reached this paper it would have been a false statement about our own amendments, in the section a referee reads most carefully. We report the aggregate as one number rather than letting a favourable half stand for the whole.

Where a specification change means two defensible analyses exist, both are computed from the same artifacts and both are reported. `dual_specification` is a required field of the verdict manifest, and a manifest that lacks it cannot be quoted; disagreement between the two specifications is a first-class finding rather than a footnote.

8.5 One identity key was deliberately relaxed

Sixteen keys define a run's provenance identity, and one — the platform ABI string — was split so that the CUDA and driver components no longer force a mismatch on their own. The reason was schedule, not principle: rented GPUs arrive with heterogeneous drivers, and an unsplit key would have refused otherwise-identical runs across machines. **We state the motive in plain words because a relaxation justified on convenience and reported as rigour is the failure this section exists to prevent**; the amendment is logged with its direction, and the remaining fifteen keys are unchanged.

8.6 The conformance surface

Pre-registration is only binding if something checks it. Every pinned quantity is a named constant that is hard-asserted before any cell is scored: the symbol set by hash, the κ grid, the five seeds, the flat headline cost of 0.0030, the evaluation sequence length, the backbone parameter count of 21,301,248 excluding the MTP heads, the $\hat{\mu}$ estimator, the training step counts, the bootstrap configuration, the causal-encoder flag, the pointwise-fine flag and the microstructure class weight. A run whose configuration differs from any of these does not produce a downgraded result; it raises and produces none.

8.7 Guards that refuse rather than decide

Two guards sit outside the decision rule, and their design constraint is that neither can change an outcome — each can only withhold one. The degeneracy guard refuses a verdict for any cell whose book is constant-direction. The power guard refuses one when a cell's across-seed range reaches the size of the effect being claimed. Both are computed after the clauses, neither mutates them, and neither can flip a survival to a null or the reverse.

That asymmetry is what made it legitimate to add them after the design was frozen: a guard that can only produce HALT_ADJUDICATE can manufacture neither a positive result nor a negative one. It is also why they are described here rather than in Section 4 — they are part of how the result is verified, not part of what is being tested. Section 4 states one asymmetry we chose not to remove and the reason.

8.8 How the defects were actually found

We keep a defect ledger, and we report it because the distribution in it is more informative than any assurance we could offer. It records 36 defects: 19 we classify as critical — meaning they would have produced a wrong published result — 15 high, and 2 medium. All but one are closed; the exception is the capacity-eviction finding itself, which is mitigated rather than closed because it is a property of the objective and not a bug.

The useful column is not severity but *how each was found*:

detection layer	defects	of which critical
Independent audit (first pass)	9	6
Builder or supervisor, outside a formal pass	8	3
Independent re-audit	6	2
Execution (found by running it)	5	2
Design review	4	2
Planted-signal canary	3	3
Independent audit (third pass)	1	1
Total	36	19

No single layer found even a third of the critical defects. The largest contributor, the first independent audit, accounts for 6 of 19. Three critical defects were found only by planting a signal of known information content and observing that it was not recovered — no review of that code would have found them, because the code was correct and the *interface* was not. Two were found only by executing the pipeline, which is the basis for the operational rule that a component whose first execution is on rented hardware is an untested component whatever its coverage says.

The reason to publish this table is that it is the strongest available argument against the verification story a paper normally tells. Code review, a green test suite and a careful author would have caught roughly a third of what mattered here. We do not claim the remaining defects are gone; we claim that seven independent detection layers were run, that each found something the others missed, and that the count of defects found by the most recent layer has not yet reached zero.

A recurring pattern is worth naming, because it changes what a reader should count as evidence. Several defects were checks that could not fail: a mutation test whose fixture made the two cases it separated numerically identical; a gate that shipped below a `return` and had never executed; a unit test whose written contract was the defect. In each, a test existed and was green. Our operating rule is therefore that a gate or negative test is suspect until it has been observed to fail, and that closing a finding requires restoring the pre-fix source in an isolated tree and watching the new test fail there.

8.9 What is released, and what a reader can check without it

The artifact is the code, the trained weights, the pre-registration with its full amendment log, the run manifests, and the scripts that regenerate every figure in this paper from those manifests at render time. No number in any figure is transcribed; each is loaded from the receipt that produced it.

That property is weaker than it sounds, and the weaker version is the one we claim. Loading from a receipt prevents a figure from contradicting the receipt it *reads*. It does not prevent a figure from reading the wrong receipt, or from reading no receipt at all for a panel whose result exists — and a generator that omits an artifact fails nothing, because there is nothing to fail. Two figures therefore carry an explicit check on top of the loading: Figure 5 asserts its drawn values against the gate's stop receipt and Figure 6 asserts its parameter counts against the checkpoints, and each has been observed to fail under a deliberately wrong constant. **The claim is thus: a figure that contradicts the artifact it loads fails to build, and where a figure could silently omit an artifact instead, we added a check rather than assumed the property.**

Two things a reader can check without rerunning anything: the anchor hash, which fixes the evaluation instrument, and Appendix E, which separates every claim in this paper into measured and estimated, with the artifact path for each. Section 7 states what remains unestablished.

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A Eviction measurement, per dimension

Table 2 gives the per-dimension point-decoder correlations summarized in Table 1. Marginal standard deviations are implied by each dimension's predict-the-mean baseline through $\mathbb{E}|X| = \sigma\sqrt{2/\pi}$, and confirm the fixture's construction: every dimension except the return is unit-variance to two decimal places, so the return is the only high-variance channel and the signal dimension is distinguished from the fillers by dependence structure alone.

Table 2: Point-decoder correlation for every fixture dimension, three seeds. Recovery of a dimension's value at bar t from bar t's own fine code. The eleven fillers are coupled by an AR(1) chain across dimensions at $\rho = 0.7993$; the signal dimension is independent of everything. Seed-to-seed variation within a class is at most 0.006, while the gap between the filler class and the signal dimension is more than 0.80.

dim	role	marginal sd	seed 0	seed 1	seed 2
0	return (high variance)	3.15	**0.9818**	**0.9815**	**0.9823**
1	filler	1.00	0.8931	0.8903	0.8920
2	filler	1.01	0.9128	0.9113	0.9109
3	filler	1.00	0.8523	0.8509	0.8565
4	filler	1.00	0.8261	0.8256	0.8319
5	filler	1.00	0.8793	0.8808	0.8816
6	filler	1.01	0.9120	0.9166	0.9101
7	filler	1.00	0.8793	0.8790	0.8738
8	filler	1.01	0.8258	0.8235	0.8242
9	signal (independent)	1.00	**0.0140**	**0.0011**	**0.0053**
10	filler	1.01	0.8513	0.8486	0.8514
11	filler	1.00	0.9179	0.9188	0.9171
12	filler	1.01	0.8969	0.8975	0.8940

Four dimensions clear a point-decoder correlation of 0.90 in each of the three seeds, which is the number the capacity arithmetic of Section 3.1 anticipates: a 10.26-bit fine budget cannot support thirteen dimensions at the fidelity that a 0.90 sign accuracy requires.

B Token-space plant: construction and calibration

The control of Section 3.4 requires a plant in token space whose information content is known exactly, is large enough to be detectable under the pre-registered branch condition (0.7–1.1 nats), and does not perturb the carrier's marginal distribution enough to be detectable by that route instead.

A first design, retired. The natural construction — a deterministic translation of one digit by an amount determined by another — cannot carry enough information on this carrier. The carrier's coarse digit marginals are already near-uniform: measured entropies of 2.3938, 2.1796 and 2.1873 nats against maxima of $\ln 11 = 2.3979$ and $\ln 9 = 2.1972$, i.e. 99.2–99.8% of maximum. Under near-uniform marginals a translation plant is capped at 0.2769 nats, below the required band, so it was retired as information-poor by construction rather than tuned.

The adopted design. The target digit is redrawn from a discretized Gaussian on the level grid, centred monotonically on the source level, with width σ_{plant} solved so that the exact mutual information lands in band. Table 3 gives the calibration receipt at solution and the same quantities recomputed over the full training stream. The closed-form and Monte-Carlo estimates agree to 9×10^{-4} at calibration and to 2×10^{-5} over the full stream; the marginal KL between the planted and carrier target distributions is 0.005 and the entropy shift is -0.005 nats, so the plant is not detectable from the marginal alone; and an unplanted control run returns a rank correlation of -8.8×10^{-5}, confirming the probe reads zero when there is nothing to read.

C Two defects in the legibility gate

Both defects are instances of the same failure — a check that cannot fail is not a check — and both were closed before the gate was relied upon. They are recorded because the gate is a hard stop on expenditure, and because the second one was material.

A skipped dimension could not lower the verdict. A dimension with too few unmasked bars to measure took a code path that continued the loop *before* the pass flag was accumulated. A gate run in which all six microstructure dimensions were too thin therefore returned `pass: True` having measured nothing, and a run with five thin and one passing dimension returned `pass: True` on a gate whose stated contract is all six. Both were reproduced before the fix. "We could not measure

Table 3: Token-space plant, calibration and full-stream receipts. The plant is solved to a target information content rather than tuned to a result; the oracle ceiling is the rank correlation an optimal predictor of the planted rule would achieve, and bounds what any model can reach.

quantity	at calibration	full stream
plant width σ_{plant}	1.0077	—
mutual information, closed form (nats)	0.9000	0.9003
mutual information, Monte-Carlo plug-in (nats)	0.9009	0.9003
|closed form − Monte-Carlo|	9.1×10^{-4}	2.0×10^{-5}
marginal KL, planted vs. carrier	0.00476	0.00480
entropy shift ΔH (nats)	−0.0054	−0.0052
oracle rank-correlation ceiling	0.926	—
unplanted control, rank correlation	-8.8×10^{-5}	—

the channel this gate exists to measure" is now a third outcome, and it halts.

The probe read one symbol of two hundred. The probe took a fixed 150,000-row head of a symbol-ordered concatenation with a contiguous train/validation cut. On the production data that window spans *one of* 200 *symbols*, so the check governing all second-stage expenditure was being made on a single instrument. The naive repair — shuffling before the split — would have broken the other property that matters, since one-minute bars are autocorrelated and an interleaved split puts near-duplicate neighbours on both sides. The fix is a per-symbol stratified split that is blocked in time *within* each symbol, which preserves both coverage and blocking.

The materiality of the two legs differs, and we measured rather than asserted it. On the production data the minimum per-dimension unmasked bar count across all 200 symbols is 277,758 against the gate's 10,000-bar floor, with zero symbols below it, so the all-thin case cannot arise here: the first defect was real but immaterial on this data. The second was severe.

D The pinned decision surface

Every quantity below was fixed before any of its inputs existed and is hard-asserted before any cell is scored. A run whose configuration differs from any of these raises rather than producing a down-graded result. The values are reproduced from the constants themselves, not from prose.

D.1 Experiment pins

constant	value	governs
`PINNED_SYMBOLS_SHA256`	`60e24f59…dc32631`	the evaluation universe
`PINNED_SEEDS`	$(0, 1, 2, 3, 4)$	five replicates, fixed up front
`PINNED_KAPPAS`	$(1.0, 1.5, 2.0, 3.0)$	the execution-filter grid
`PINNED_HEADLINE_COST`	0.0030	flat round trip, the pessimistic band end
`PINNED_MONEY_SEQ_LEN`	512	eval surface and training context
`PINNED_BACKBONE_PARAMS`	21,301,248	backbone *excluding* the MTP heads
`PINNED_STEPS_STAGE1`	26,003	tokenizer budget, matched across arms
`PINNED_STEPS_STAGE2`	26,003	backbone budget, matched across arms
`PINNED_MU_ESTIMATOR`	`expectation`	conditional mean, not mode
`PINNED_ENCODER_CAUSAL`	`True`	forced for every cell
`PINNED_FINE_POINTWISE`	`True`	the re-specified interface
`PINNED_MICRO_POINT_WEIGHT`	3.0	the class weight λ
`PINNED_MICRO_LEGIBILITY_MIN`	0.90	the gate, on every live micro dimension
`PINNED_CODEBOOK_MIN_UTILIZATION`	0.95	FSQ-scoped collapse check; the BSQ cells report and do not gate

D.2 Decision thresholds and the deflation recipe

constant	value	governs
ECON_FLOOR_IR	0.5	clause 4 materiality floor
DSR_N_TRIALS	60	clause 5 trial count
DSR_THRESHOLD	0.95	clause 5 deflated-Sharpe threshold
trial factorization	cells $\times$ horizons $\times$ κ	$5 \times 3 \times 4 = 60$
DSR_HORIZONS	$(5, 15, 60)$	the horizon axis
DSR_VAR_SR_BASIS_CELL	5	dispersion basis is the *placebo* arm
de-annualization horizon	15	the unit SR_0 is compared against
PLACEBO_DISPERSION_TRIPWIRE	1.5×	disclose if the placebo arm is over-dispersed
PINNED_BOOT	$B = 10{,}000$, seed 20260704, $\alpha = 0.05$	the paired bootstrap
BOOT_POWER	0.80	the MDE's power term
FRAC_NEG_BAND	$[0.05, 0.95]$	degeneracy guard, sign-balance leg

Two conventions, stated so a reader is not left to infer them. DSR_N_TRIALS is a **pin**, not a derived quantity: its value 60 equals the product of the factorization (5 cells $\times$ 3 horizons $\times$ 4 values of κ), and conformance asserts both the literal and the factorization, but the decision to enumerate trials that way was made by us before the run and is not a measurement. Appendix E classifies it accordingly. And FRAC_NEG_BAND governs only the sign-balance leg of the degeneracy guard; the filter-binding leg of the same guard tests the exact endpoints 0 and 1 with no band, so an activity of 0.999 passes where a fraction-negative of 0.999 halts. **That asymmetry is endpoint-only by ruling, not by omission.** It was adjudicated against both a proposal to band the leg to match its sibling and a proposal to leave it, and the leg was kept as is: activity at κ^* is a scale-dependent fraction rather than a count, and it is not a sufficient statistic for whether the filter did per-bar work, since a cell can read interior at κ^* while being fully inert at three of the four grid values. Three compensating disclosures are required in its place: activity_decisions_by_kappa across the full grid rather than the κ^* scalar alone, the trade *count* beside the fraction, and $\mathrm{std}(\hat{\mu})/\theta$ at κ^*. Section 7.5 states what the ruling leaves uncovered.

Two of these carry a history that Section 8.4 summarizes and the amendment log states in full. The trial count was reduced from 180 to 60 when seeds were recognized as replicates rather than configurations, and the dispersion basis was moved to the placebo arm because the earlier all-arms basis made clause 5 anti-correlated with its own hypothesis. The first tightens the criterion and the second loosens it; the net is indeterminate before the run, and Section 8.4 explains why we report it that way rather than as a single direction.

D.3 Evaluation geometry

quantity	value
window	2021-01-01 to 2025-01-01, one-minute bars
train/forward split	2023-10-20, the 0.7 point
forward blocks	6; block 0 selects κ and is excluded
headline region	blocks 1–5, $\approx$ 2024-01-01 to 2025-01-01
periods per instrument at $h = 15$	35,063
scored instruments	40 of the 200 in the lake
effective breadth	1.5
label look-forward	60 bars
serial-correlation margin	60 bars
leading embargo	120 bars

D.4 Content addressing and environment

The Gate-A anchor is a fixed evaluation of a frozen checkpoint through the full harness, re-proven twice from unpiped commands before and after any change to the evaluation instrument.

artifact	value
anchor results hash	`sha256:3f86882a63dd06c78…`
anchor causal sweep	420 of 420 checks
anchor dataset hash	`sha256:1d3ec4f8db6686e2…`
anchor tokenizer hash	`sha256:ce7c8c4ae0b62c89…`
anchor predictor hash	`sha256:6d118ede6bfcab2e…`
universe Merkle root	`sha256:5dfd667d05b97bda…`
lake	7,024 Parquet files, 200 symbols, 304,625,181 bars
Python	3.11.7
NumPy	2.4.6
PyTorch	2.12.1

The environment is pinned by a lockfile and load-bearing results are re-verified under it. The amendment log carries 58 distinct revision tags, each with its date, its reason, and the direction it moves the criterion, recorded before the amendment was evaluated.

D.5 Slot-to-field bindings for §6

Every empty slot in Section 6 binds to exactly one named field. The binding is recorded here so that filling §6 after the run is a mechanical substitution rather than a judgement.

These bindings were re-verified against the emitter on 2026-08-11 and six of the twenty-one were wrong as first written. The clause outcomes were given as `.passed` where the emitter writes `.pass`; the power-guard fields were given names the emitter does not use; and the legibility gate was given a verdict-manifest path when it is a *separate* Stage-1 receipt that no verdict manifest contains. The corrected table is below. We report the error rather than silently fixing it, because D.5's own claim — that no slot has more than one defensible source — is worth less if the table naming the sources was not itself checked.

§6 slot	field	artifact
per-dimension sign accuracy	per_dim.<7..12>.sign_acc	Stage-1 legibility receipt, *not* the verdict manifest
minimum across the six	min(per_dim.*.sign_acc)	
gate outcome	pass	
clause 1 outcome	clauses.1_paired_ci.pass	verdict manifest
clause 2 outcome	clauses.2_mde_paired.pass	
clause 3 outcome	clauses.3_placebo_validity.pass	
clause 4 outcome	clauses.4_economic_floor.pass	
clause 5 outcome	clauses.5_dsr.pass	
degeneracy guard	verdict.halted_for_degeneracy	
power guard	verdict.halted_for_power	
primary outcome	verdict.primary	
emitted outcome	verdict.emitted	
failing clauses	verdict.failing_clauses	
dual-spec pair	dual_specification. {v1_2_original,v1_5_amended} .primary	
specifications agree	dual_specification.agree	
net IR, seed mean	point_diagnostics / cell_ir	
across-seed range	power_guard.per_cell.<c>. ir_range_across_seeds	
decision activity at κ^*	degeneracy_guard.per_cell.<c>. activity_decisions_seed_mean	
seed scatter (Fig 10)	power_guard.per_cell.<c>.ir_by_seed	
claimed effect (Fig 10)	power_guard.checks.<name>. {delta_ir_claimed, worst_within_cell_ir_range}	
abstract verdict slot	verdict.emitted	

Two notes a reader needs. The manifest key for the minimum detectable effect is clauses.2_mde_paired.mde_paired; mde_paired_scoring_only is the *attribute* on the bootstrap object that supplies it and appears under no key. And the outcome words a manifest can carry are SURVIVES, NULL and HALT_ADJUDICATE only — INCONCLUSIVE is an adjudication reached against the manifest under the trigger of Section 6.3, never a value read from it. The activity row is published whatever it reads, including at an endpoint, for the reason Section 7.5 gives.

D.6 Required fields of the verdict manifest

A manifest missing any field below is refused rather than reported, and a paper quoting a manifest that lacks one is quoting an artifact the harness would not have accepted.

required field	why it is required
primary	the outcome word; absent means no verdict was reached
clauses.*.{rule,passed,value}	the rule string is persisted, not restated in prose
dual_specification	both analyses from the same artifacts; disagreement is a finding
degeneracy_guard.{armed,halted}	armed is computed, so it cannot assert inspection it did not perform
degeneracy_guard.per_cell.*	per-seed values, so an adjudication sees the distribution
power_guard.{armed,halted}	same contract as the degeneracy guard
mu_diag.{frac_negative, activity_decisions}	the guards' inputs; unreadable inputs HALT rather than pass
external-validation disclosure	the BSQ sentence of Section 7.2, verbatim
conformance.*	every pin in D.1 and D.2, re-asserted at scoring time

Two of these are worth naming as design choices rather than bookkeeping. armed is a *measurement* of whether the guard could read its inputs, not a literal — an earlier contract let a guard report armed: true having inspected nothing, which is the check-that-cannot-fail pattern sitting in the decision path. And the external-validation disclosure is required rather than optional precisely because it is the sentence an author would most like to omit.

Ruling of record for D.2: `docs/degeneracy_guard_activity_leg_RULING.md`, dated 2026-08-11, carried as pre-registration amendment tag v1.6.28.

E Claims audit

Every quantitative claim in this paper carries an inline comment in the LaTeX source naming the artifact it came from — 99 such comments across 47 distinct artifact paths, stripped from the submitted version and present in the released source. This appendix summarizes them by status.

Four statuses are used, and the boundaries between them are the point of the table. **Measured** means a value read from a committed receipt produced by executing something. **Derived** means computed in closed form from measured inputs, with the derivation shown at the point of use. **Pin** means a constant we chose before seeing data — a pin is not a finding, and no pin should ever be read as evidence. **Pending** means the quantity does not exist until the pre-registered run completes.

E.1 The mechanism (§3)

quantity	value	status	source
per-dim reconstruction, return channel	0.98	measured	interface re-spec receipt
per-dim reconstruction, correlated block	0.82–0.92	measured	same, dims 1–8, 10–12
per-dim reconstruction, independent dim	0.001–0.014	measured	same, dim 9, three seeds
fixture entropy match, per-dim deficit	0.4631 → 0.462	measured	filler calibration
feature-space plant, information	1.151 nats	derived	$\frac{1}{2}\ln(1+c^2)$ at $c = 3$
feature-space plant, probe correlation	< 0.05 at all 40 evals	measured	canary v6 receipt
token-space plant, information	0.900 nats	measured	token-control receipt
token-space plant, recovery	94.4%	derived	0.8496/0.90027
per-bar legibility, before (fixture)	0.5112	measured	re-spec gate record, `old`
per-bar legibility, token-control probe	0.5135	measured	planted carrier, *not* the fixture
chance rate	0.50	pin	probe construction
re-spec ALONE, 3 seeds	0.4995–0.5182	measured	re-spec gate record, **pass false**
with $\lambda = 3$ class weight	0.9047 mean, 0.8839 min	measured	λ-search; restated rule
legibility gate threshold	0.90	pin	conformance
bits per token, both designs	20.058	derived	$\sum \log_2 L_i$
class weight λ	3.0	pin	fitted on the fixture, then pinned

Two entries deserve a reader's attention. The 1.151 nats is a closed-form quantity computed from a receipt constant rather than a measurement, and the derivation is shown in Section 3.3 rather than asserted. And $\lambda = 3.0$ is a pin whose value was *fitted on the synthetic fixture*; Section 7.11 states the transfer risk and names the gate that guards it.

E.2 The data (§5)

quantity	value	status	source
bars	304,625,181	measured	lake surface check
instruments	200	measured	same
symbol-months present	7,024 of 9,600	measured	lake enumeration
OHLCV-usable	303,283,972 (99.56%)	measured	ingest DQ table
microstructure-usable	300,311,536 (98.58%)	measured	same
microstructure availability	99.0%, none below 0.7	measured	same
stale / zero-volume share	0.175% / 0.376%	measured	same
delisted at ingest scan	41	measured	universe bootstrap
data ends inside the window	17	measured	lake enumeration
warm-up drop, thinnest instrument	809 of 279,930	measured	DQ spot-check
z-score half-lives	1,440 / 5,760	pin	feature config
clip range	[−5, +5]	pin	feature config
feature dimensions, total / live	16 / 13	measured	constants, mask census

E.3 The design (§4)

quantity	value	status	source
minimum detectable effect, $h = 15$	3.518 annualized IR	derived	MDE inputs receipt
economic floor	0.5	pin	verdict constants
prior, |RankIC| at $h = 5$ / $h = 15$	0.0270 / 0.0197	measured	M2 univariate screen
DSR trial count	60	pin	conformance; $= 5 \times 3 \times 4$, but the enumeration was our choice, not a finding
bits-per-bar gap, FSQ vs BSQ	0.058	derived	vocabulary arithmetic
realized predictor params, FSQ / BSQ	31,795,200 / 31,725,568	measured	both checkpoints
of which MTP heads (identical)	10,493,952	measured	same; 33.0% of the model
backbone excluding MTP	21,301,248 / 21,231,616	measured	same; the pinned figure
parameter gap	69,632 (0.219% of model)	measured	realized counts
signed-return ACF at lag 60	0.0067 mean, 0.0191 worst	measured	embargo premise
absolute-return ACF at lag 60	0.2323 mean, 0.3604 worst	measured	same
placebo: micro–price correlation	0.2037 → 0.0005	measured	placebo mechanism doc
placebo: micro lag-1 autocorrelation	0.4165 → 0.0006	measured	same
capacity handicap in IR units	—	**not quantified**	Section 7.3
filter activity at κ^*, real data	0.0193–0.4702	measured	three cell-1 money artifacts
break-even cost, $h = 15$	0.00232–0.02082%	measured	horizon break-even receipt
on the pre-registered grid	$-\infty$	measured	same; the grid floor *is* the fee

Two rows deserve a reader's attention. The capacity handicap is disclosed and unquantified — no cell in the pre-registration brackets it. And the last three rows were carried as **pending** through several drafts, on the assertion that no artifact recorded them; all three are measured, in the three cell-1 money artifacts, and the correction is recorded at Section 7.5 and Section 7.6 rather than quietly applied. The $-\infty$ is not a missing value: it is what the pre-registered instrument returns, and Section 7.6 is about why.

E.4 Verification (§8)

quantity	value	status	source
defects in the ledger	36 (19 critical)	measured	defect ledger
largest single detection layer	6 of 19 critical	derived	ledger tabulation
determinism penalty	13.175%	measured	CUDA probe report
false bit-exactness claims	49 of 49 records	measured	the D20 entry
same-seed divergence	0.5662 vs 0.99883	measured	h-sweep vs money-leg
gross IR vs own-lean benchmark, Δ	+0.9972–+3.4355	measured	skill-vs-bias receipts
clearing zero, one-sided α=.05	2 of 3	measured	seed 2 +1.5574, seed 4 +0.0454
seed 4, t of that Δ	1.6971	derived	0.9972/0.5876
seed 4, two-sided 95% low / p	−0.1545 / 0.0897	derived	stricter than pre-registered
cross-build replay agreement	2.59×10^{-4}	measured	exact-run control arm
always-long IR of the sample	+0.3704	measured	same; the deflationary alternative
amendment log	58 revision tags	measured	pre-registration
anchor causal sweep	420 of 420	measured	Gate-A manifest
test suite	947 passing	measured	`pytest` at `78a502d`

E.4a A header-less tag is cited on a shipped clause

Appendix F.1 records that the amendment log carries 58 distinct revision tags across 55 top-level entries, three tags — v1.0, v1.6.24 and v1.6.27 — appearing as in-body references without headers of their own. One consequence belongs here rather than there, because it touches a field that ships.

The verdict manifest cites one of those header-less tags on its headline clause. The `rule` string of `4_economic_floor` — a field written into every emitted manifest — ends "see `placebo_capacity_disclosure` and §7 v1.6.27".

The pointer resolves, and the distinction matters: header-less is not absent. The tag marks a full correction in the body of the log, recorded there as the clause it replaced being false. It sits inside the entry that introduced the placebo capacity disclosure the same rule string names, so a reader following the pointer arrives at the disclosure and its correction together.

v1.6.24 is the weaker case. It appears in the log only as two back-references, with no entry of its own defining it, and is cited from several places outside the pre-registration — including this paper, a limitations document, a run manifest, and the repository's own instructions. A reader following that tag into the log finds it used but never defined.

The entries are not backfilled, and the reason is the ruling. Retro-dating pre-registration entries to tidy a count would make the log less trustworthy, not more: a log whose entries can appear after the fact is not a pre-registration. The gap is therefore disclosed rather than closed — which is the same disposition, and the same reasoning, as the activity-leg ruling in Appendix D.2. Ruled by the supervisor, 2026-08-11.

E.5 Everything in §6

The gate receipt of Section 6.1 — twelve sign accuracies, their base rates and the two gate outcomes — is **measured**, as is the λ sweep of Section 6.4. What remains **pending** is everything the ablation would have produced: no verdict, no cell comparison, no secondary diagnostic. Those bindings are tabulated in Appendix D.5 and named beside each slot in the source, so filling them would be mechanical substitution rather than judgement. And Section 6.3 no longer carries all five interpretation branches: the outcome selected one, and the other four were deleted rather than softened, exactly as pre-committed.